

Landing force reveals new form of motion-induced sound camouflage in a wild predator

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Abstract

Predator-prey arms races have led to the evolution of finely-tuned disguise strategies. While the theoretical benefits of predator camouflage are well established, no study has yet been able to quantify its consequences for hunting success in natural conditions. We used high-resolution movement data to quantify how barn owls (*Tyto alba*) conceal their approach when using a sit-and-wait strategy. We hypothesized that hunting barn owls would modulate their landing force, potentially reducing noise levels in the vicinity of prey. Analysing 87,957 landings by 163 individuals equipped with GPS tags and accelerometers, we show that barn owls reduce their landing force as they approach their prey, and that landing force predicts the success of the following hunting attempt. Landing force also varied with the substrate, being lowest on man-made poles in field boundaries. The physical environment therefore affects the capacity for sound camouflage, providing an unexpected link between predator-prey interactions and land-use. Finally, hunting strike forces in barn owls were the highest recorded in any bird, relative to body mass, highlighting the range of selective pressures that act on landings and the capacity of these predators to modulate their landing force. Overall, our results provide the first measurements of landing force in a wild setting, revealing a new form of motion-induced sound camouflage and its link to hunting success.

eLife assessment

This **fundamental** work substantially advances our understanding of animals' foraging behaviour by monitoring the movement and body posture of barn owls in high resolution and assessing their foraging success. With a large dataset, the evidence supporting the main conclusions is **compelling**. This work provides new corroboration for motion-induced sound camouflage and has broad implications for understanding predator-prey interactions.

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Introduction

Predation represents one of the strongest forms of selection in nature (1–6). As a result, animals have evolved sophisticated adaptations to modify the sensory information they emit (4,7–10). Camouflage has been widely studied as an anti-predator defence, with mechanisms including background matching, disruption, and self-shadow concealment facilitating predator avoidance (11–14). Predators also show adaptations to reduce detection by prey e.g. in their colour, markings and/or behaviour (15–17). However, in general, predator camouflage is far less understood due to the challenges of simulating predation in controlled settings (18) and observing predation attempts in the wild (19). This has hindered our understanding of the evolutionary forces driving predator camouflage and explains why predator cues have yet to be linked to prey capture success.

Predation typically requires movements of a predator towards its prey, either during a pursuit or an ambush, which usually exposes chasing predators to detection (15,20). Indeed, motion makes individuals more conspicuous (5,12,21,22). But motion also produces sound through the generation of vibrations and turbulence (23,24), which can be detected by prey with acute hearing. Many predators alter their movements accordingly, for instance, by moving slowly during the pursuit, which may provide both acoustic and visual camouflage (8), particularly when combined with a background colour matching (15,20,25). While the direct link to hunting success remains unclear (15,26), selection should favour camouflage strategies that reduce sound emission in quiet environments. The resulting arms race may explain why many nocturnal species have acute senses of hearing, which they rely on to detect danger or prey (8,27,28).

The silent flight of owls is one of the most iconic examples of noise camouflage. Quiet flight is achieved through comb-like serrations on the leading edge of owls' wing feathers that break up the turbulent air and minimize associated sound production (8,29). This should provide advantages when hunting on the wing. However, most owls also launch attacks from perches, which involves moving from one perch to the next as they approach their prey (30–32). Landing also produces vibrations, and hence sound, with the intensity being proportional to the landing force (33). In this dynamic sit-and-wait strategy, landing likely becomes a key element of prey approach. We use high-frequency GPS and accelerometer data to investigate the landing dynamics of this sit-and-wait strategy in wild barn owls (*Tyto alba*). Specifically, we quantify whether the landing force varies with (i) the time until the hunting strike (i.e. hunting motivation), (ii) perch type (i.e. environmental context), and (iii) body mass, which varies between males and females (34). Finally, we test the extent to which the magnitude of the landing force affects hunting success.

Results

We used GPS loggers and accelerometers to record high resolution movement data during two consecutive breeding seasons (May to August in 2019 and 2020) from 163 wild barn owls (79 males and 84 females) breeding in nest boxes across a 1,000 km² intensive agricultural landscape in the western Swiss plateau. Of these individuals, 142 belonged to pairs for which data were recovered from both partners (71 pairs in total, 40 in 2019, 31 in 2020). The remaining 21 individuals belonged to pairs with data from one partner (11 females and 1 male in 2019; 4 females and 5 males in 2020).

Measurement of landing force

We used the acceleration data to identify 84,855 landings. These were further categorized into perching events ($n = 56,874$) and hunting strikes ($n = 27,981$), depending whether barn owls were landing on a perch or attempting to strike prey on the ground (**Fig. 1A and B**, see methods for specific details on behavioural classification). We extracted the peak vectorial sum of the raw acceleration during each landing and converted this to ground reaction force (hereafter “landing force”, in Newtons) using measurements of individual body mass (see methods for detailed description).

Hunting strikes had landing forces over four times higher than perching events (**Fig. 1C**, **Tables S2**; ratio: 4.5, z-ratio: 486.3, $p < 0.001$). When converted to multiples of body weight, hunting strikes had peak forces that were equivalent to approximately 13 times body weight, whereas perching events involved forces roughly three times body weight.

Determinants of landing force

We conducted two sets of analyses to investigate factors that influence the variation in landing force in different landing contexts: perching events and hunting strikes.

Barn owls employing a sit-and-wait strategy land on multiple perches before initiating an attack, with successive landings reducing the distance to the target prey (**Fig. 2C**). We analysed the landing forces involved in sequences of perching events in relation to perch type (poles, buildings, and trees: identified using GPS data) and the time before an attack (i.e. pre-hunt time: an indication of hunting motivation). The most important predictor of landing force in perching events was perch type. Perching events on buildings were associated with the highest forces (8.96 N, CI: 8.90 – 9.01 N; **Tables S3**), closely followed by landing on trees (8.86 N, CI: 8.81 – 8.90 N; **Tables S3**). Poles were associated with the lowest landing force (8.33 N, CI: 8.28 – 8.38 N, **Tables S3**).

Importantly, within perch types, there was a reduction in landing force with time until the next hunting attempt, with the pattern differing with perch type ($\text{EDF}_{\text{poles}} = 4.22$, $p < 0.001$; $\text{EDF}_{\text{buildings}} = 1.00$, $p < 0.001$; $\text{EDF}_{\text{trees}} = 1.50$, $p = 0.005$; **Fig. 2A**, **Tables S3**; $n_{\text{tot}} = 40,306$ perching events; see **Fig. S4** for the full representation and derivative plot). When barn owls perched on poles, the landing force showed a marked decrease in the last 30 minutes before the hunting strike, whereas landing force only showed a marginal linear reduction with time before strike for landings on buildings (**Fig. 2A**, **Fig. S4**). Landing force did not show any significant reduction with time for perching events on trees (**Fig. 2A**, **Fig. S4**, **Tables S3**). Our analysis also revealed a clear temporal pattern in the birds’ use of perch types: owls launched more attacks from poles than from trees, with the fewest attacks launched from buildings (**Fig. 2B**). The pattern of variation in landing force according to perch type and hunting motivation, and the pattern of perch use, were consistent for both males and females, despite females consistently exhibiting greater landing forces than males (**Tables S3**).

Additionally, our analysis of hunting strike force showed that both hunting strategy and success were related to strike force (**Tables S4**). When hunting on the wing, successful strikes involved greater forces than unsuccessful strikes ($n_{\text{tot}} = 24,464$; successful strikes: $n_{\text{succ}} = 5,830$, 40.3 N, CI: 39.5 – 41.2 N; unsuccessful strikes: $n_{\text{unsucc}} = 18,634$, 38.4 N, CI: 37.7 – 39.2 N). This was not the case when barn owls hunted from a perch ($n_{\text{tot}} = 3,517$; successful strikes: $n_{\text{succ}} = 1,042$, 38.8 N, CI: 37.7 – 40.0 N; unsuccessful strikes: $n_{\text{unsucc}} = 2,475$, 38.5 N, CI: 37.6 – 39.5 N).

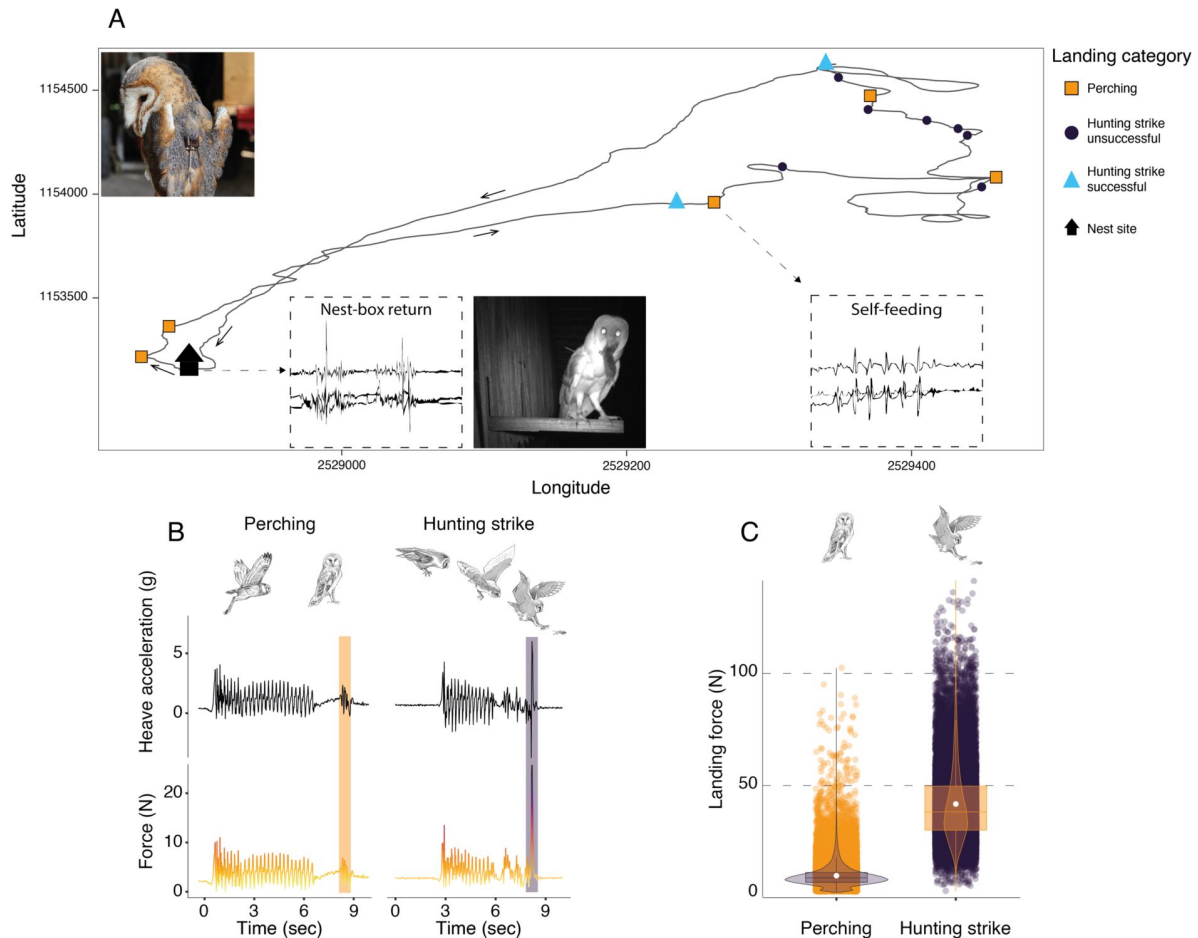


Figure 1.

Sequence of perching and hunting strike events throughout a Barn owl foraging trip.

A) GPS tracks (black line) during one complete foraging trip (duration = 73 minutes) performed by a female barn owl, with perching events (squares), unsuccessful (circles) and successful (triangles) hunting strikes. Black arrows indicate the flight direction. Successful hunting strikes were identified by the presence of self-feeding events (identified from the acceleration data), or by the direct return to the nest box (identified from the acceleration data and validated with the nest box camera footage). Inset panels show an example of the tri-axial acceleration signals corresponding to both nest-box return and self-feeding behaviours (see [Fig S3](#) for detailed representations). B) The heave acceleration and the associated force during a perching event (highlighted in orange) and a hunting strike (highlighted in dark purple). C) Variation in peak landing force for perching events (orange dots, $n = 56,874$) and hunting strikes (dark dots, $n = 27,981$). White dots show the estimated mean, and data distribution is represented by both violin and box plots. The owl picture at the top left of Panel A is courtesy of J. Bierer, and owl drawings are courtesy of L. Willenegger, all used with permission.

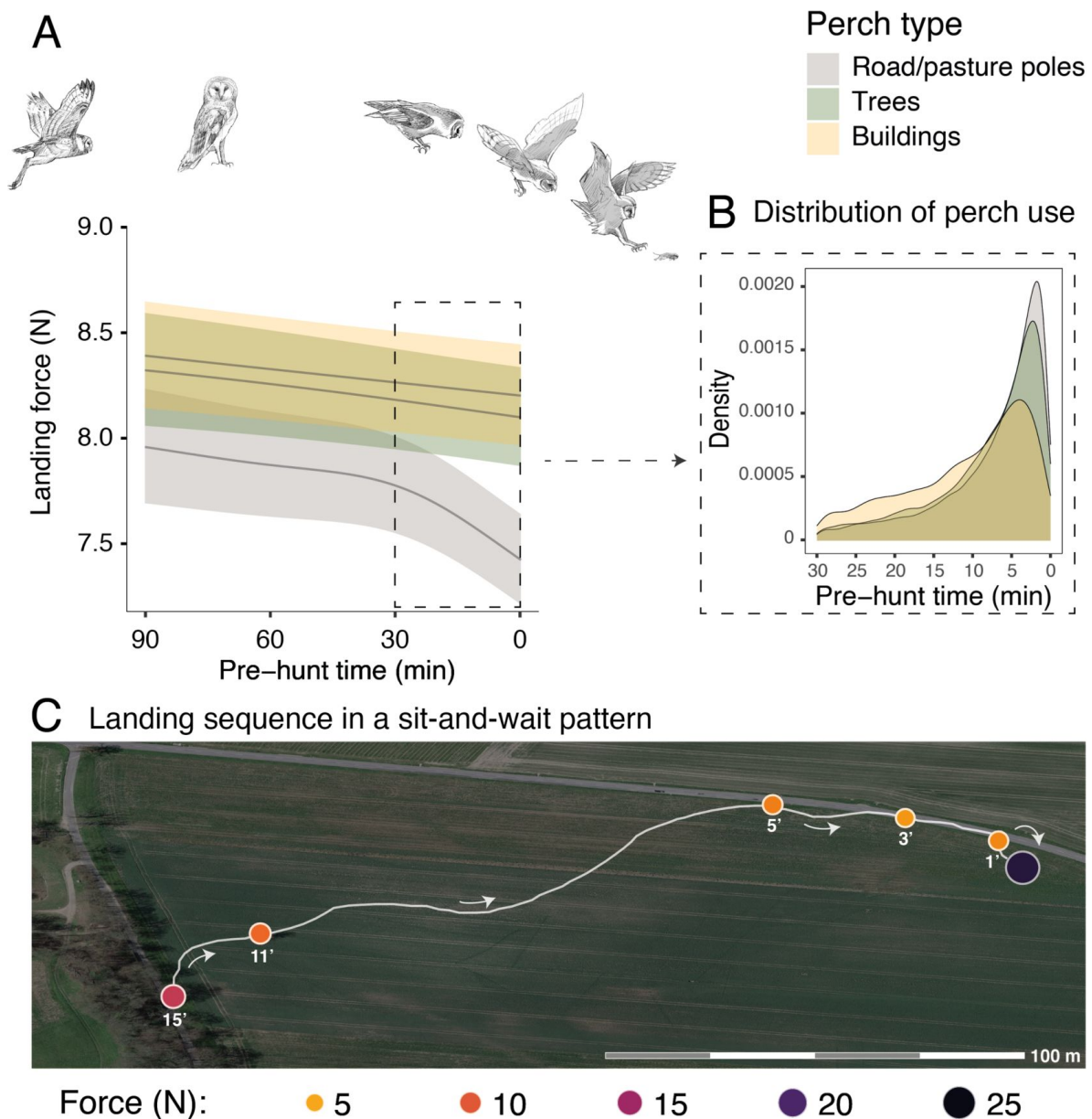


Figure 2.

Sequential changes in perch type and landing force prior to hunting strikes during a sit-and-wait hunt

A) Landing force during perching events in relation to the time until the next hunting event and perch type. Each line represents the predicted mean for each perch type (here shown for males), with the 95% confidence intervals. B) The selection of perch type in relation to time until the next hunting strike, highlighting the change in perch type that occurs ~10 minutes prior to a strike. C) A sequence of perching events (orange circles) prior to a successful strike (purple circle) for a typical sit-and-wait hunt, showing the variation in peak landing force through time. White arrows indicate flight direction and numbers under each perching event indicate the time until the next hunting attempt (i.e. pre-hunt time). The owl illustrations at the top of Panel A are courtesy of L. Willenegger, used with permission.

Sexual dimorphism and foraging behaviour

Sexual dimorphism in body mass was marked among our sampled individuals. Males were lighter than females (84 females, average body mass: 322 ± 22.6 g; 79 males, average body mass 281 ± 16.5 g, **Fig S6**). Males also displayed higher nightly hunting effort than females (Males: 46 ± 16 hunting attempts per night, $n=79$; Females: 25 ± 11 hunting attempts per night, $n=84$; **Fig. 3A**, **Fig S8**). However, females were more likely to use a sit-and-wait strategy than males (females: $24\% \pm 15\%$, males: $13\% \pm 10\%$, **Fig.S9**). As a result, the number of perching events per night was similar between males and females (Females: 76 ± 23 perching events per night; Males: 69 ± 20 perching events per night; **Fig S8**).

We conducted two different analyses to assess whether hunting strategies differed in success and efficiency (i.e. foraging trip duration). To assess the influence of strategy on foraging trip duration, we extracted the number of sit-and-wait hunting attempts and divided this by the total number of hunting attempts and analysed this in relation to the trip duration (min) and sex. Our analysis showed that trip duration increased with the use of the sit-and-wait strategy (**Fig. S11**, **Table S8**): barn owls that only used the sit-and-wait strategy (sit-and-wait frequency =1) took an average of 15 minutes longer to provide prey to the nest than those that only hunted on the wing (sit-and-wait frequency = 0). Nonetheless, barn owls were more successful when using a sit-and-wait strategy, with success also varying with sex. Males were more successful than females, both for the sit-and-wait strategy (males: 34.5%, CI: 31.6% – 37.5%; females: 26.8%, CI: 24.4% – 29.2%, **Fig. 3D**, **Tables S5**) and hunting on the wing (males: 26.1%, CI: 24.7% – 27.6%; females: 19.1%, CI: 17.7% – 20.5%, **Fig. 3D**, **Tables S5**).

Landing force also varied with sex, with females generating landing forces that were 26% higher than males on average during perching events (**Tables S2**, **Fig. 3C**) (females: 9.94 N, CI: 9.63 N – 10.27 N; males: 7.91 N, CI: 7.65 N – 8.18 N; ratio F/M: 1.26, 95% CI: 1.2–1.31; **Fig. 3C**). Males and females had similar landing forces during hunting strikes, with females generating forces that were only 6% higher than males on average (females: 40.8 N, CI: 39.49 N – 42.18 N; males: 38.41 N, CI: 37.14 N – 39.71 N; ratio F/M: 1.06, 95% CI: 1.01–1.11; **Fig S10**). However, when considered per unit of body mass (see methods), males exhibited lower forces than females when perching (males: 28.2 N/Kg, CI: 27.4 N/Kg – 29.0 N/Kg; females: 30.9 N/Kg, CI: 30.1 N/Kg – 31.8 N/Kg; **Fig S10**), but higher forces than females in hunting strikes (males: 136.2 N/kg, CI: 132.3 N/Kg – 140.2 N/Kg; females: 126.4 N/kg, CI: 122.8 N/Kg – 130.1 N/Kg, **Fig S10**).

Additionally, we performed two additional analyses to investigate potential variation in landing force and flight speed between sexes. We extracted the median ground speed (in m s^{-1}) of each flight prior to each hunting attempt and analysed this in relation to the sex of the individual. This showed that males flew slightly more slowly than females when searching for prey on the wing. Males flew slower than females by 0.23 ms^{-1} (Average flight speed males: 5.24 ms^{-1} , CI: 5.15 ms^{-1} – 5.33 ms^{-1} ; average flight speed females: 5.47 ms^{-1} , CI: 5.38 ms^{-1} – 5.56 ms^{-1} , **Fig. 3B**, **Table S7**).

Pre-hunt landing force predicts hunting success for sit-and-wait strategy

Finally, we analysed whether the landing force in the last perching event before each hunting attempt (i.e. pre-hunt perching force) predicted variation in hunting success. Our results showed that hunting strategy was the strongest predictor of success (**Fig 3**, **Table S6**, $n = 3,040$ hunting strikes from 151 individuals, see methods for details on data filtering). When hunting from the wing, the force applied during pre-hunt perching events had no effect on hunting success (**Fig. 4**, **Tables S6**, odds ratio: 1.07, CI: 0.97 – 1.17, $p = 0.19$). However, during sit-and-wait hunts, where the distance between the last perch and the prey is rather short (median distance 6.5 m, **Fig.S5**), pre-hunt perching force predicted hunting success (**Fig.4**, **Tables S6**). When barn

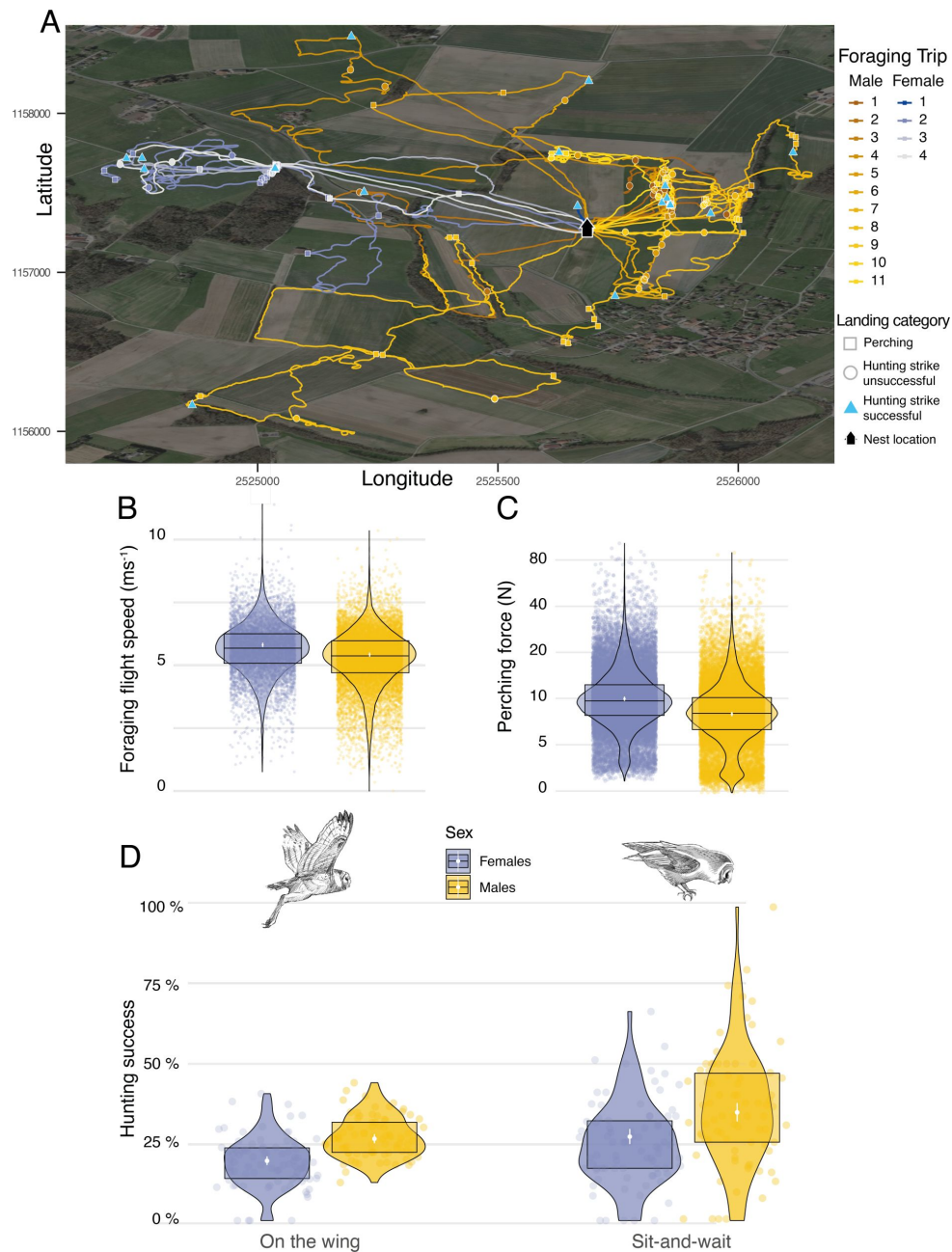


Figure 3

Sexual differences in foraging behaviour, landing force and hunting success.

A) GPS tracks showing the foraging activities of a barn owl breeding pair during one complete night. Movement patterns of both male (yellow lines) and female (blue lines) are shown, with colour scale changing from the first trip of the night (foraging trip 1) to the last one (Male: $n_{\text{max}} = 11$; Female: $n_{\text{max}} = 4$). Perching events (squares), unsuccessful (circle) and successful (triangle) hunting attempts are shown for each foraging trip. B) Variation in foraging flight speed for female (blue dots, $n = 9,223$) and male (orange dots, $n = 18,019$) barn owls ($n_{\text{females}} = 84$; $n_{\text{males}} = 78$). C) Variation in peak landing force during perching events (perching force) for female (blue dots, $n = 30,378$) and male (yellow dots, $n = 26,496$) barn owls. D) Variation in hunting success when barn owls hunted on the wing or used the sit-and-wait strategy for female (blue dots, $n_{\text{on-the-wing}} = 8,136$, $n_{\text{sit-and-wait}} = 1,981$) and male (yellow dots, $n_{\text{on-the-wing}} = 16,328$, $n_{\text{sit-and-wait}} = 1,532$) barn owls. For visualization purposes, each dot shows the average hunting success of each individual for both hunting strategies. Panels B to C) White dots and bars show the mean and the 95% CI around the mean respectively, and data distribution is represented by both violin and boxplots. Owl drawings are courtesy of L. Willenegger, all used with permission.

owls hunted directly from a perch, the chance of success decreased by 15% for every 1 N increase in pre-hunt perching force (odds ratio: 0.85, CI: 0.79 – 0.99, $p = 0.04$). Perch type and wind speed were dropped from the final model after model selection (**Table S6b**).

Discussion

Silent flight is considered crucial for owls hunting on the wing (29). But these predators also use a sit-and-wait strategy, with owls in this study achieving greater hunting success when launching attacks from a perch. Here, owls typically approach prey by moving between multiple perches. They must therefore avoid detection both in flight and as they land, as the benefits of silent flight may be negated if owls are detected during touchdown. We found that barn owls hunting from pasture poles reduced their landing force as they got closer to their prey. This suggests that soft landings are a novel form of acoustic camouflage, with predators reducing their motion-induced sound production in response to information on prey presence, which they gather as they move between perches in the final phase of the hunt. The landing force also affected the success of the subsequent strike, demonstrating the link between the predator camouflage and hunting success. However, the relatively low R-squared value (**Table S6a**) suggests that hunting success is affected by additional factors such as prey behaviour, substrate type and grass length (35).

Owls appeared to vary their perch use in relation to their motivation to hunt. For instance, barn owls that landed on a perch 30 to 90 minutes before a strike may have done so without the immediate intention of hunting. This phase was associated with greater use of buildings and trees, which are much higher structures than poles. Owls may therefore preferentially use these perches to rest or gather information over a wider area. The shift in predominant perch type to pasture poles 5-10 minutes before hunting strikes appears to represent a shift to periods of active prey searching. In some systems, the choice to hunt from pasture poles may be driven by prey availability, as pasture poles may be embedded in hedgerows or area with longer grass where prey density might be higher. However, this is unlikely to be the case in our system as pasture poles generally occur in the middle of short-cropped grass (**Figure S2**). Instead, poles may offer advantages in being close to the ground, enhancing opportunities for owls to refine their estimates of prey location, prey type or size. Indeed, the opportunity to gather information from perches could help explain the greater overall hunting success in attacks launched from a perch, compared to hunting on the wing.

In birds, landings are primarily governed by the need to maintain flight control and minimize the risk of injury (36–38). For instance, Harris' hawks (*Parabuteo unicinctus*) landing in controlled conditions postponed the stall until they were as close to the landing perch as possible (36). Such a strategy could serve two functions in barn owls using sit-and-wait hunting; minimizing both the energy dissipated on impact and the associated sound production (33). This raises the question of why owls would ever land with anything above the minimal force. To date, almost all studies have examined landings in controlled conditions (36,39), yet in the wild, birds are faced with a range of perch types and landing conditions. Perch characteristics are likely to play a pivotal role, as forces tend to be absorbed to a greater degree by compliant substrates (40). In support of this, landings on buildings were associated with the highest mean forces, and higher forces than tree branches (**Table S5**), which would be more compliant, with the extent varying with branch type and diameter (38). It was therefore notable that forces were lowest for landings on poles, which, like buildings, are rigid. However, poles occur in open habitat (**Figure S2**), providing a predictable landing surface that can be approached from all directions, facilitating control through optimal use of the wind vector. Landing force may therefore be influenced by the access options as well as the substrate type. There may also be greater incentive to reduce landing force on poles, since they are close to the ground and sound attenuates with distance (29,41,42).

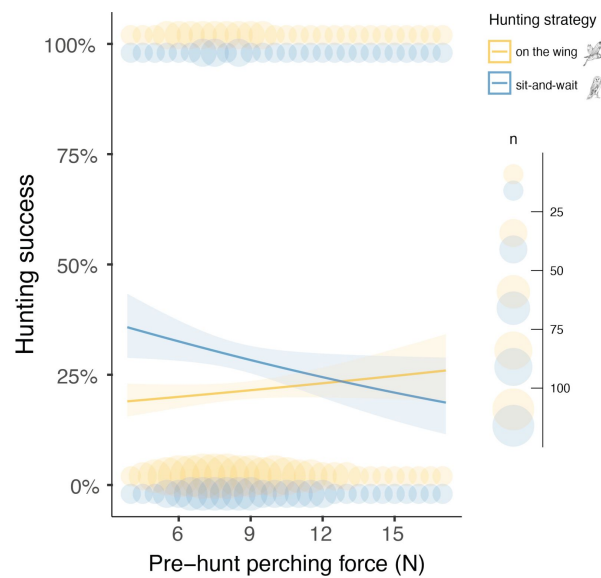


Figure 4.

Pre-hunt perching force affects hunting success during sit-and-wait hunting.

Variation in hunting success according to the pre-hunt perching force (N), depending on whether owls hunted on the wing (yellow) or using the sit-and-wait strategy (blue). Solid lines show the estimated means (averaged over both sexes), and the shaded area corresponds to the 95% confidence intervals around each mean. Blue and yellow circles show the force, recorded during the last perching event before hunting (pre-hunt perching), pooled to the nearest integer N value for representation purposes, when hunting on the wing or using the sit-and-wait strategy, respectively. Circle size is related to the amount of data with the same value. The owl illustrations at the top right are courtesy of L. Willenegger, used with permission.

The biggest difference in landing force was observed between perching events and hunting strikes. Strike forces in our study are the highest recorded in any bird, relative to body mass, with maximal force reaching more than 34 times the bodyweight (100N). This exceeds estimates previously reported for captive barn owls (43), and the kicking strike of the secretary bird (*Sagittarius serpentarius*) that reached an average of 5.1 times body weight (44). Unlike secretary birds, whose kicking strength depends solely on the muscular power of their lower limbs, owls use the dynamics of their entire body in flight. While this likely minimizes the chances of prey escape, it is also associated with a potential risk of injury (36,45). Our results likely underestimate the true peak forces, as acceleration was recorded at 50 Hz (for reference, data on force development in controlled car crashes are typically recorded at > 2 kHz). Nonetheless, our data can still provide new insight on the selective pressures that have influenced owl morphology. Indeed, the lower limbs of owls allow for the dual function of absorbing shock during pre-hunt perching and generating extremely powerful hunting strikes.

We find that males and females had very similar strike forces, despite their substantial difference in body mass. This indicates that there might be a selective pressure for a minimum strike force, which males may generate by increasing or maintaining their flight speed prior to a strike to a greater extent than females. Males had a lower flight speed during prey searching, most likely due to their lower body mass (46). While the difference in flight speed was relatively small, slower flight could still have advantages in (i) providing additional time to localize prey, and (ii) enabling birds to manoeuvre into the strike phase (47,48). This may help explain why males have higher success when hunting on the wing. Males also showed greater hunting success than females in the sit-and-wait strategy. Here, a lower body mass could also provide advantages by facilitating lower impact, and hence quieter landings.

Given that sit-and-wait hunting is associated with higher success, why do male barn owls not use this strategy more (it was associated with less than 10% of hunting attempts)? Male barn owls engage in intense hunting activity in the breeding season, providing over 15 prey per night in our study. Our results showed that foraging duration increases with the use of the sit-and-wait strategy. Thus, the time required to capture prey appears to be the key element influencing the choice of hunting strategy in males. Females provide fewer prey items and the additional time required for sit-and-wait hunting may therefore be less of a constraint. Furthermore, sit-and-wait hunting may require less flight time and hence effort, which is likely to be particularly advantageous for females due to their greater body mass (and higher flight costs per unit mass).

In conclusion, we use high-frequency movement data to propose a novel form of acoustic camouflage and demonstrate that the magnitude of predator cues can influence hunting success (49). Minimizing landing force, and associated sound production is likely to be particularly pertinent for nocturnal predators, which operate in quiet environments and target prey with an acute sense of hearing (27,28,50). Importantly, the ability to minimize landing force was modulated by the perch characteristics, providing a potential link between landing impact and habitat characteristics. This suggests there could be spatial patterns in the effectiveness of acoustic camouflage and, ultimately, hunting success. The availability of different perch types could therefore be an additional, and previously unrecognized, aspect of habitat and territory quality, and, in this case, one that is strongly linked to land-use practices.

Methods

Study area and tag deployment

Data were collected from wild barn owls breeding in nest boxes across the Western Swiss plateau, an area of 1,000 km² characterized by open and largely intensive agricultural landscape (51). Over 380 nest boxes were checked for barn owl clutches between March and August in 2019 and

2020, following Frey and colleagues' protocol (52). During the two breeding seasons, 163 breeding barn owls (84 females; 79 males) were equipped with data-loggers (2019: 43 males and 49 females; 2020: 36 males and 35 females, **Fig.S1**).

Adult barn owls were captured at their nest sites approximately 25 days after the first egg hatched using automatic sliding traps that are activated when birds enter the nest box. AXY-Trek Mini loggers (Technosmart, Italy) were attached as backpacks (**Fig.1A**) using a Spectra ribbon harness (Bally Ribbon Mills, USA). These units include a GPS, set to record animal location at 1 Hz, 30 minutes before sunset until 30 minutes after sunrise, to get the full nightly activity period. The loggers also include a tri-axial accelerometer, which recorded acceleration continuously at 50 Hz (recording range $\pm 16g$, 10-bit resolution). After 10 days (± 2 days), loggers were recovered by recapturing adult barn owls at their nest sites, again using automatic sliding traps, with data recorded for 5 nights on average (± 1 night). Owls were weighed at both visits and the averaged body mass from the two measurements was used for later analysis. Each device weighed on average 12.4 ± 0.1 g, which corresponds on average to 4% of the barn owl's total body mass (min = 3%, max = 5%, female average body mass: 322 ± 22.6 g; males average body mass 281 ± 16.5 g) and therefore never exceeded the limit of 5% of the bird's body mass (53).

In parallel to each logger deployment, motion sensitive camera traps (Reconyx HC500 hyperfire, resolution of 3.1 megapixel) were positioned at the entrance of all nest boxes to document when animals returned to the nest with prey (**Fig.1A**). Camera traps were scheduled to record burst of three pictures when motion was detected. Moreover, wind data were collected using portable weather stations (Vantage Vue, Davis Instruments corp.) mounted 2.0 m from the ground (standard anemometer measurement height) within 100 m of each nest. Wind speed and direction were recorded every 10 minutes.

Behavioural classification

We used Boolean-based algorithms (54) to classify flight, landing, hunting strikes, and self-feeding from the onboard acceleration and GPS data (see below). Behaviours were summarized in one-second intervals and linked to the closest GPS location in time. Flight, hunting, and self-feeding behaviours were ground-truthed using video footage of 2 captive barn owls equipped with the same data loggers. Further validations were undertaken for hunting behaviour (detailed below).

Behavioural classifications used the raw acceleration data, the vectorial dynamic body acceleration (VeDBA) (a summary metric of body motion) and body pitch angle (**Fig.S3**, **Table S1**). VeDBA was derived by smoothing the raw acceleration data over 0.5 s (the period of two complete wingbeat cycles), to estimate the static/gravitational component, subtracting this from the raw acceleration in each of the three acceleration channels (55), and calculating the vectorial sum from the resulting "dynamic" components. Pitch angle was derived using the arcsine of the static acceleration in the heave axis (56,57) and smoothed over 1 second.

Flights were identifiable from the acceleration data as periods of take-off, travelling and landing (**Fig.S3A**). Take-offs were characterized by a switch from a standing to a horizontal posture (Δ pitch angle $> -10^\circ$) and high-amplitude VeDBA (> 1 g) (43). Travelling flight was associated with smoothed VeDBA values > 0.1 g, and body pitch values $< 30^\circ$. Finally, landings were identifiable as changes from low to high pitch angles (Δ pitch angle $> 10^\circ$) and a typical final spike in all three acceleration axes (VeDBA > 1 g). Periods that did not correspond to flight were categorized as stationary behaviour.

Landings were further classified as either perching event, where owls landed on a perch prior to a hunting attempt or hunting strikes/prey capture attempts (**Fig.1**). Landing types were categorized using the rate of change in pitch angle (strikes: Δ pitch angle $> 6^\circ$) and the amplitude of the peak acceleration (strikes: Δ VeDBA > 1.3 g) generated by the impact with the prey/ground,

which were both much greater for hunting strikes than perching events (**Fig.S3B** [↗](#)). Hunting strikes were classified using the Boolean-based classification algorithm (**Table S1** [↗](#)), whereas perching events were identified as the termination of flights that did not end with a hunting strike.

Owls hunt to provision themselves and their offspring. Self-feeding was evident from multiple and regular acceleration peaks in the surge and heave axes (resulting in peaks in VeDBA values > 0.2 g and < 0.9 g, **Fig.S3D** [↗](#)), with each peak corresponding to the movement of the head as the prey was swallowed whole. Prey provisioning events were identified from variations in the sway, corresponding to the owl walking inside the nest box (**Fig.S3C** [↗](#)). Both start and end phases of the nest box visits were characterized by a rapid change in the pitch angle (enter: Δ pitch angle $< -1.5^\circ$; exit: $< 0.5^\circ$) along with an increase in the heave and VeDBA values (enter: Δ VeDBA > 0.5 g; exit: Δ VeDBA < -0.9), as owls leapt in/out of the nest box. Successful provisioning hunts were further confirmed using nest box camera data when available and, in all cases, by manually checking that the GPS data matched the nest site to identify cases where the owls returned with a single prey for their offspring. Unsuccessful strikes were therefore inferred from identified hunting strikes that were not followed by a provisioning to the nest and/or self-feeding event (**Fig.1A** [↗](#)).

Data processing

Data from the onboard accelerometers can be used to estimate landing force during perching and hunting strikes (**Fig.1B** [↗](#)), as force is equal to the product of mass and acceleration. To estimate landing force, we extracted the peak vertical component of the ground reaction force in Newtons (N) for every landing event, taking the maximum value of the vectorial sum of the raw acceleration (in units of gravitational acceleration, g), multiplying this by the body mass of the bird (in kg) (58 [↗](#), 59 [↗](#)).

Hunting strikes were categorized according to whether owls hunted on the wing or from a perch to assess factors affecting the landing force of perching events involved in the sit-and-wait strategy. We therefore considered that owls were using the sit-and-wait strategy if they flew for a maximum of 1 second before the strike (corresponding to c.a. 6.5 m from the last perch). Hunting on the wing was defined as cases when birds flew for at least 5 seconds prior to the strike (c.a. 81.7 m from the last perch). Hunting strikes that did not fit into either category (8% of all hunting strikes) were excluded from the dataset. When barn owls were hunting on the wing, we also estimated foraging flight speed by extracting the median ground speed (in ms^{-1}) over the last 20 seconds preceding each hunting strikes.

Finally, perch type was estimated by extracting the median location of each perching event. The habitat within 2 m was then classified according to the main perch type available: trees, roadsides, and pasture poles (hereafter referenced as “poles”), and buildings, and assigned as the perch type for each perching event. Habitat categories (roads, settlements, single trees, forest) were provided by the Swiss TLM3d catalogue (Swiss Topographic Landscape Model, resolution 1-3 m depending on the habitat feature) and habitat data were provided by the “Direction générale de l’agriculture, de la viticulture et des affaires vétérinaires (DGAV)” and the “Direction des institutions, de l’agriculture et des forêts (DIAF)”, for states of Vaud and Fribourg respectively.

Statistical analyses

We first assessed how landing force varied between hunting strikes and perching events, before evaluating the factors that explained variation within each category. This excluded perching events made when owls were loaded with prey, where the landing force will likely be influenced by the extra mass carried.

We fitted a linear mixed model (LMM) of the landing force (log-transformed) where fixed factors included the landing context (a two-level factor: hunting strike or perching event), the sex of the individual (a two-level factor: Female and Male) and their interaction. Sex was included in the

model to control for sexual differences in foraging strategy as well as a sexual dimorphism in body mass (31 [↗](#)). The model included bird ID as a random intercept to account for repeated measurements of the same individual over multiple nights, and night ID (nested in bird ID) to account for repeated measurement of the same individual within the same night. The same random effect structure was applied to all the following LMs and GLMs as they were fitted to dataset of similar grouping structure. We also fitted a LMM of the landing force during hunting strikes (log-transformed). Fixed factors in the model included hunting success (a two-level factor: successful and unsuccessful), the hunting strategy (a two-level factor: perching and flying) and their interaction. Sex was also included as a fixed factor.

We next fitted a generalized additive mixed-effects model (GAMM) to assess how the landing force (log-transformed) varied between perching events. Specifically, we examined whether this was affected by the physical environment (perch type, wind), or motivation (owls can perch for long periods between hunts, and the most pertinent currency determining landing force may therefore vary between periods of resting and active searching). Time until the next hunting strike was extracted for every perching event and included as a continuous fixed covariable in the model. An interaction between a smoothed function of the time until the next hunting strike and perch type was also included, using a thin plate regression spline and the “by” condition, with the number of bases per smooth term (k) set at a conservative value of 9. The sex of the individual, windspeed and perch type (a three-level factor: pole, tree, building) were included as linear predictors in the model. The model included the random intercept effect of bird ID (included with $bs="re"$ in a smooth function).

Our GAMM of landing force showed that owls perched more softly the closer they came to the next hunting strike. To identify periods when there was a significant change in landing force, we calculated the first derivative $f'(x)$ of the estimated smoothed relationship between the time to the next strike and the peak landing force, according to each perch type, to highlight significant periods of positive or negative relationships (60 [↗](#), 61 [↗](#)). Periods of significant change were identified as those time points where the simultaneous confidence interval on the first derivative does not include zero.

Finally, we performed a set of analyses to investigate how hunting success varied with sex and hunting strategy, and most specifically whether success might be influenced by the landing force involved during perching events. To study how hunting success overall varied with sex and hunting strategy, we ran a first generalized linear mixed-effect model (GLMM) with hunting success as binary response variable (1 = successful, 0 = unsuccessful). In this first model, the sex of each individual, the hunting strategy and their interaction were included as fixed effects.

Then, we fitted a second GLMM with hunting success as a binary response variable to specifically investigate whether the landing force applied in the last perching events would influence the success of the following hunting attempts. Hypothesizing that landing force might affects barn owl detectability, we only selected hunting strikes that were immediately preceded by a perching event (hereafter pre-hunt perching). We also selected hunting strikes that occurred < 90 seconds after the last perching event to maximize the probability of capturing a response to the pre-hunt perching force. The threshold of 90 seconds corresponded to the lower tercile of the distribution of time differences between perching and hunting strikes. The fixed effects included in this second model were pre-hunt perching force (i.e the force applied during perching events directly preceding each hunting attempt), hunting strategy and their interaction. The sex of the individual, windspeed and the interaction between sex and hunting strategy were also included as fixed effects in the model.

In birds, body mass usually influences flight speed (46 [↗](#)). We therefore hypothesized that the sexual dimorphism in body mass present in barn owls' population might influence the speed at which males and females would fly when foraging on the wing. This could in turn impact on their

ability to locate and target prey on the ground and therefore ultimately influence hunting success when hunting on the wing. To test this hypothesis, we fitted a second LMM with foraging flight speed as continuous response variable and the sex of each individual as fixed effect.

Finally, we fitted a LMM to assess how preferences of a given hunting strategy might affect barn owls foraging trip duration. The model included the foraging trip duration (min) as response variable. The model also included the frequency of use of the sit-and-wait strategy (number of hunting attempts in the sit-and-wait per trip divided by the total number of hunting attempts per trip), the total number of hunting attempts per trip, and the sex as predictors.

All statistical analyses were conducted with R 4.0.5 (R Core Team, Vienna, Austria), with RStudio (RStudio Team, 2020) as graphic user interface. LMMs and GLMMs were fitted with the functions *lmer* and *glmer*, respectively, implemented in the package ‘lme4’ (R package v1.1-27.1) (62) and we used the package ‘lmerTest’ (R package v3.1-3) (63) to estimate p-values. GAMM model was fit using the *gam* function from the package ‘mgvc’ (R package v1.8-34) (64–66). For all models, linear predictors were centered and scaled to mean zero and units of standard deviation (i.e., z-scores) to ensure comparability among variables. We selected the optimal structure of the fixed component of each models using a multi-model selection framework ranking the selected models according to the Akaike information criterion (67,68), using an automated stepwise model selection procedure in which models are fitted through repeated evaluation of modified calls extracted from the model containing all the meaningful variables, corrected for small sample sizes (AICc) (69). The final models were chosen as the best models among the candidate models within $\Delta AICc < 2$, that was always relatively low (between 1 and 4) (see selection model tables that are reported in Supplementary Materials). Additionally, we performed pairwise comparisons using the *emmeans* function from the package ‘emmeans’ (R package v1.6.0) (70) to further assess differences between predictors level. Models were fitted, checked for collinearity between predictors and assumptions were verified by visually inspecting residual diagnostic plots. Descriptive statistics are reported as Mean $\pm$ SD, unless specified otherwise.

Ethical statements

This study meets the legal requirements of capturing, handling, and attaching Animal tracking devices to barn owls in Switzerland from the Department of the consumer and veterinary affairs (legal authorizations: VD, FR and BE 3213 and 3571; capture and ringing permissions from the Federal Office for the Environment). The 5% weight limit of the GPS-ACC loggers was considered acceptable because of the short period during which the tracking devices were deployed on birds. Tagged birds survived and reproduced with similar annual fitness output than non-tagged individuals.

Data availability

The datasets and codes generated and/or analysed during the current study will be available from the date of publication without restrictions on request.

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Authors Contributions

K.S.: conceptualization, methodology, software, investigation, formal analysis, data curation, writing–original draft, writing–review and editing, and visualization. E.M.: investigation, methodology, writing–review and editing. R.S., R.B. and S.P.: methodology, writing–review and editing. B.A.: methodology, writing–review and editing, funding acquisition. P.B.: methodology, investigation, formal analysis, writing– original draft, writing–review and editing, supervision. A.R: supervision, writing–review, project administration and funding acquisition. E.L.C.: conceptualization, methodology, investigation, formal analysis, writing–original draft, writing–review and editing, supervision.

Competing interests

The authors declare no competing interests.

Supplementary information

Figure S1

: GPS tracks (in black) of the 163 breeding barn owls used in this study recorded in 2019 and 2020 in western Switzerland.

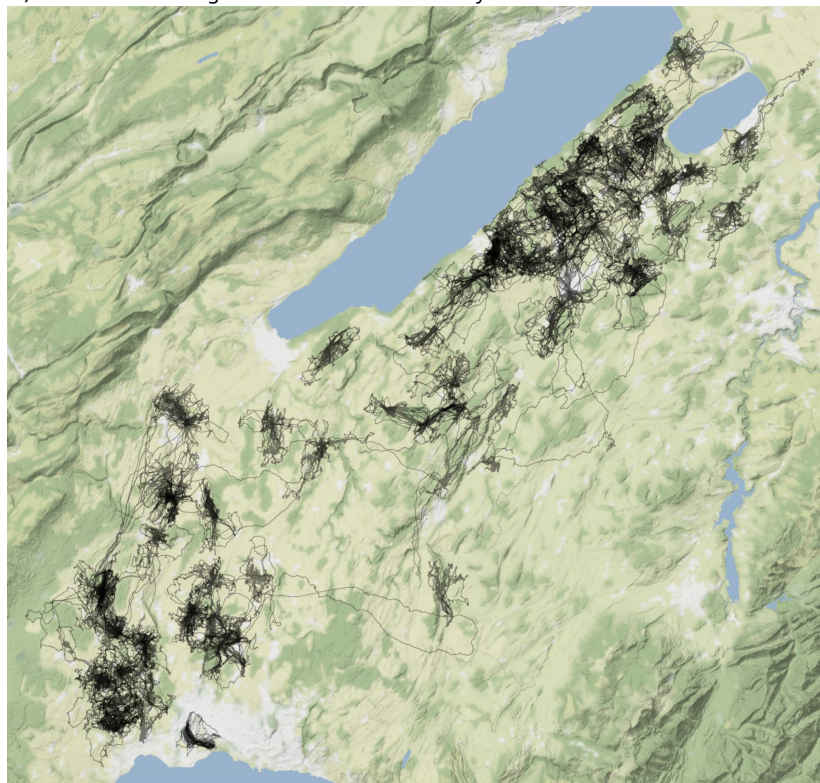


Figure S2

Typical Barn owl foraging ground and nest location in western Switzerland.

A) Example of how pasture poles are usually located within the agricultural landscape which represents the main habitat for barn owls in western Switzerland. B) Typical barn in which nest boxes are usually installed on the Swiss plateau.

A



B



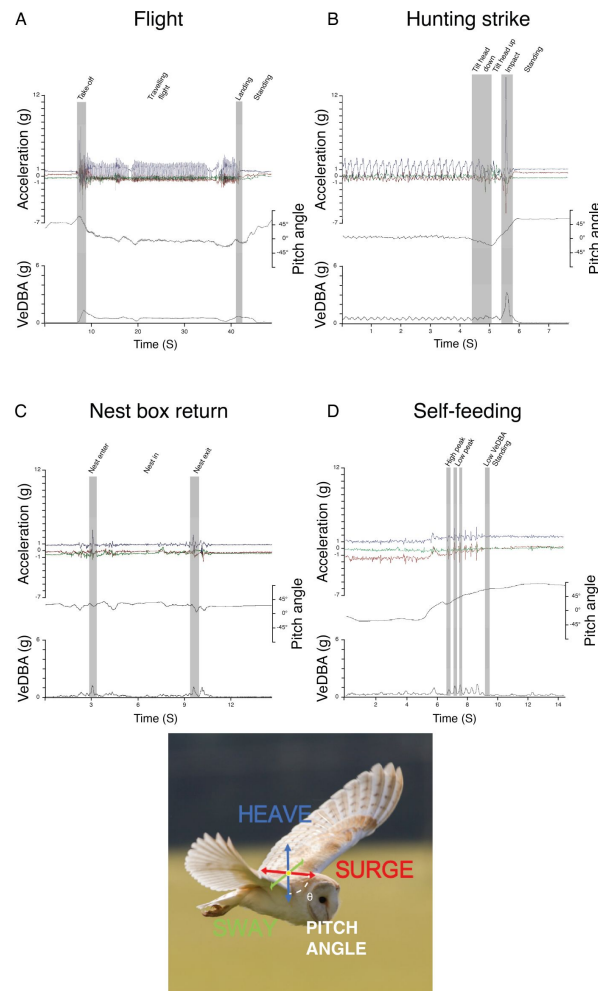


Figure S3

Behaviour classification from accelerometer data.

Time series data of the different behaviour classified using Boolean approach showing changes in the raw tri-axial acceleration, body pitch angle, and the vectorial sum of the dynamic body acceleration (VeDBA) corresponding to A) flight, B) hunting strikes, C) nest box visit and D) self-feeding events. Behaviour specific base element used in the Boolean classification are shown in grey bands. Note that hunting strikes involves greater acceleration amplitude, VeDBA and body pitch variation between landings in context of usual perching (here at the end of flight sequence).

Figure S4

Complete variation in landing force according to pre-hunt time.

Graphic representation of (A) the complete variation of the landing force calculated during perching and (B) the corresponding first derivative events in relation to the time (hours) until the next hunting event depending on whether owls perched on poles (in grey), trees (in green) and buildings (in yellow). Each line represents the predicted means for each perch type (averaged over male individuals), and shades show the 95% confidence intervals around each mean.

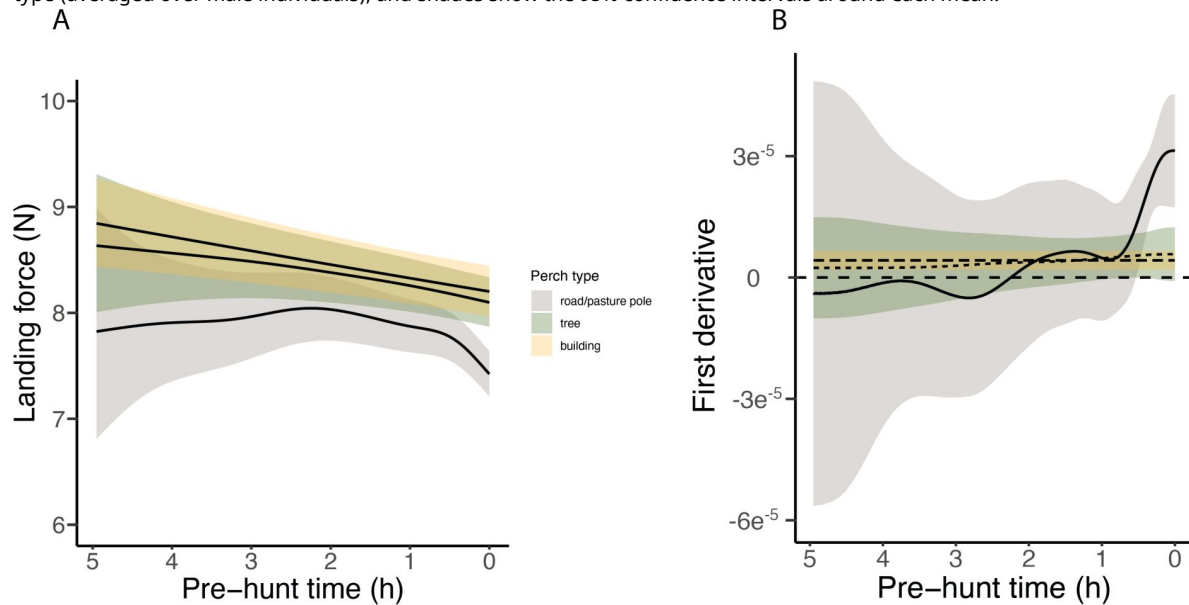
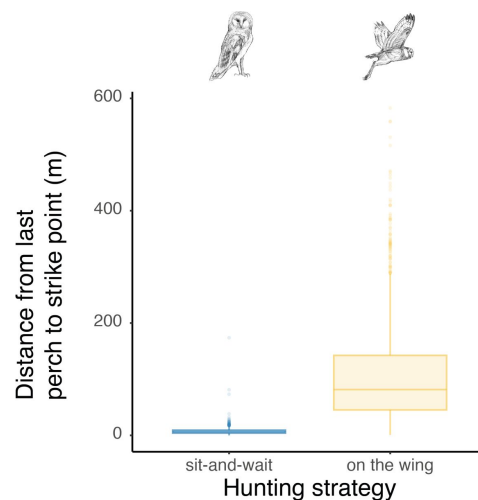


Figure S5

Differences in distance between the last perch and strike location according to the hunting strategy.

Box plots of the variation in distance between perching location and hunting strikes among both perching and flying hunting strategies, showing a significant difference in distance between perch and strike location according to the hunting strategy (Wilcoxon test: $W = 2344881$, $p\text{-value} < 0.001$). Boxes boundaries highlight the first and the third quartile of the range distribution of the data. The line within each box marks the median and whiskers above and below boxes indicate the 10th and 90th percentiles.



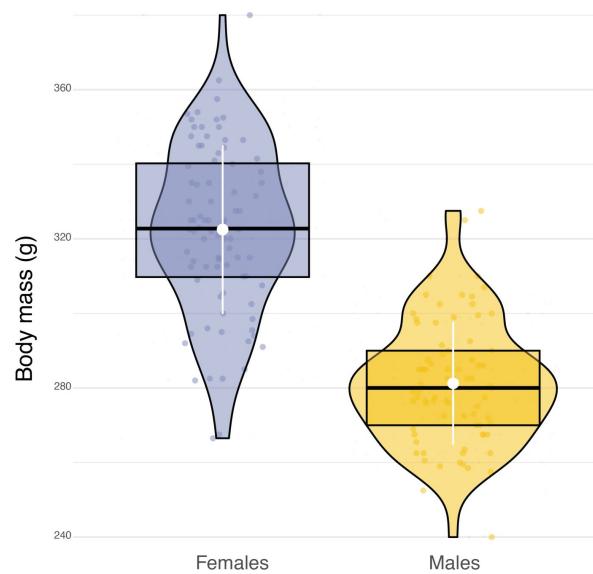


Fig. S6.

Sexual dimorphism in body mass.

Box plots of the variation in body mass between females (blue dots, $n = 84$) and males (orange dots, $n = 79$). White dots and bars respectively highlight the average and the standard deviation of body mass.

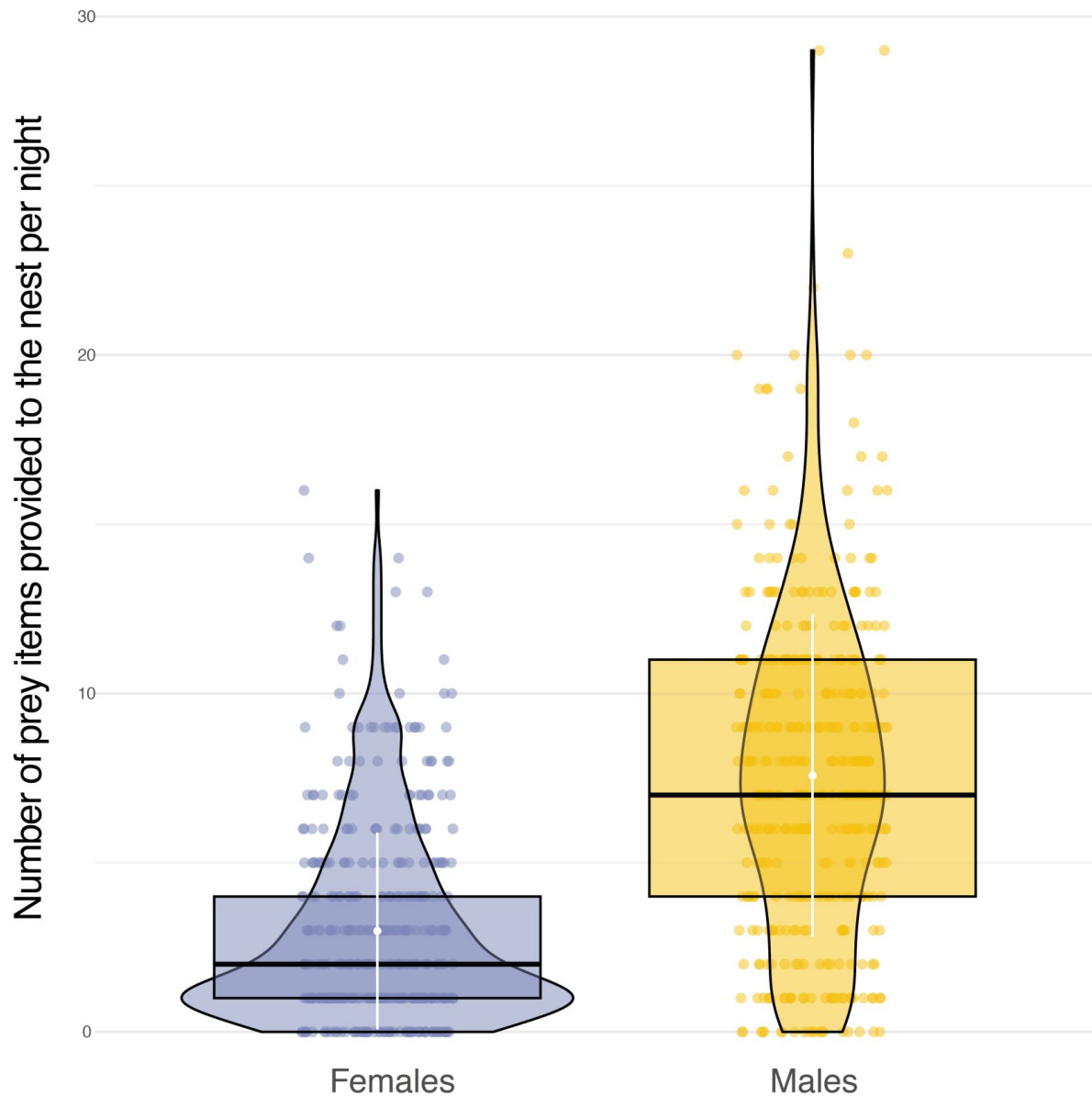


Figure S7.

Sexual differences in food provisioning.

Box plots of the variation in the number of prey items delivered to the nest each night between females (blue dots, $n = 1,226$) and males (orange dots, $n = 3,105$). White dots and bars respectively highlight the average and the standard deviation of the number of prey items delivered to the nest each night.

Figure S8.

Sexual differences in hunting activity.

Box plots of the variation in the number of hunting strikes and perching events performed each night by females (blue dots, nb perching events = 22,134, nb hunting strikes = 8176) and males (orange dots, nb perching events = 19,657, nb hunting strikes = 15,158). White dots and bars respectively highlight the average, and the standard deviation of the number of hunting strikes and perching events performed each night.

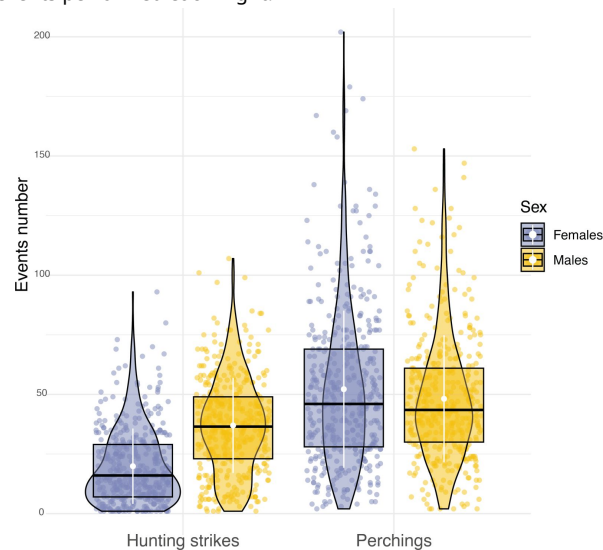
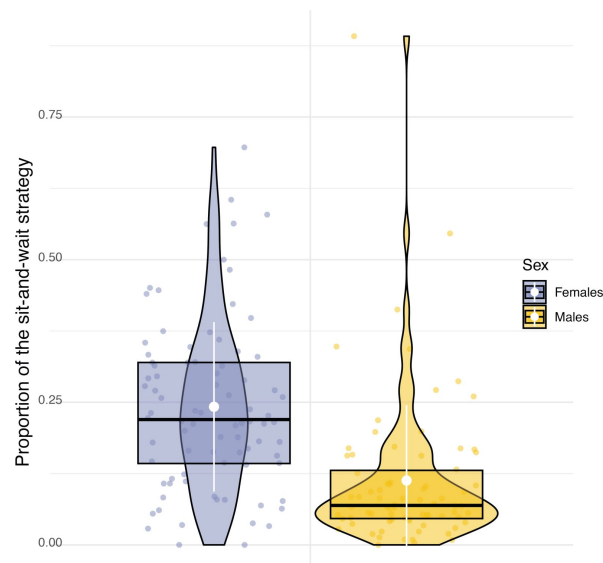


Figure S9.

Sexual difference in the use of hunting strategy.

Box plots of the variation in the proportion of use of the sit-and-wait strategy for females (blue dots) and males (orange dots). White dots and bars respectively highlight the average and the standard deviation of the proportion of use of the sit-and-wait strategy.



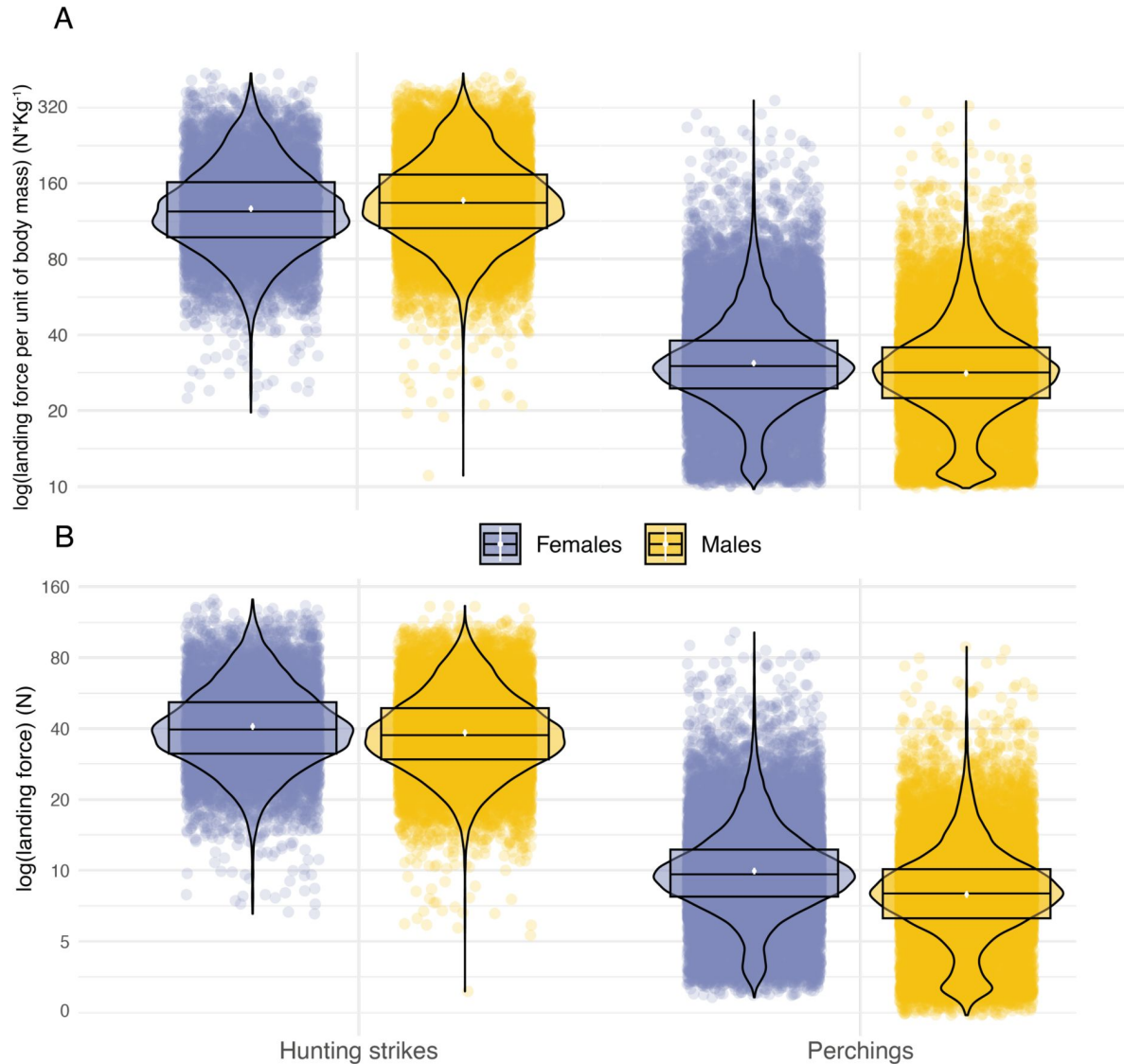


Figure S10.

Sexual comparison of landing force as a function of landing context.

Variation in peak landing force, on the log scale, involved in perching events and hunting strikes between female (blue dots, nb hunting strikes = 10,117, nb perching events = 30,378) and male (orange dots, nb hunting strikes = 17,864, nb perching events = 26,496) individuals. A) Shows landing force per unit of body mass and B) shows variation in peak landing force. White dots show the estimated mean, and data distribution is represented by both violin and box plots.

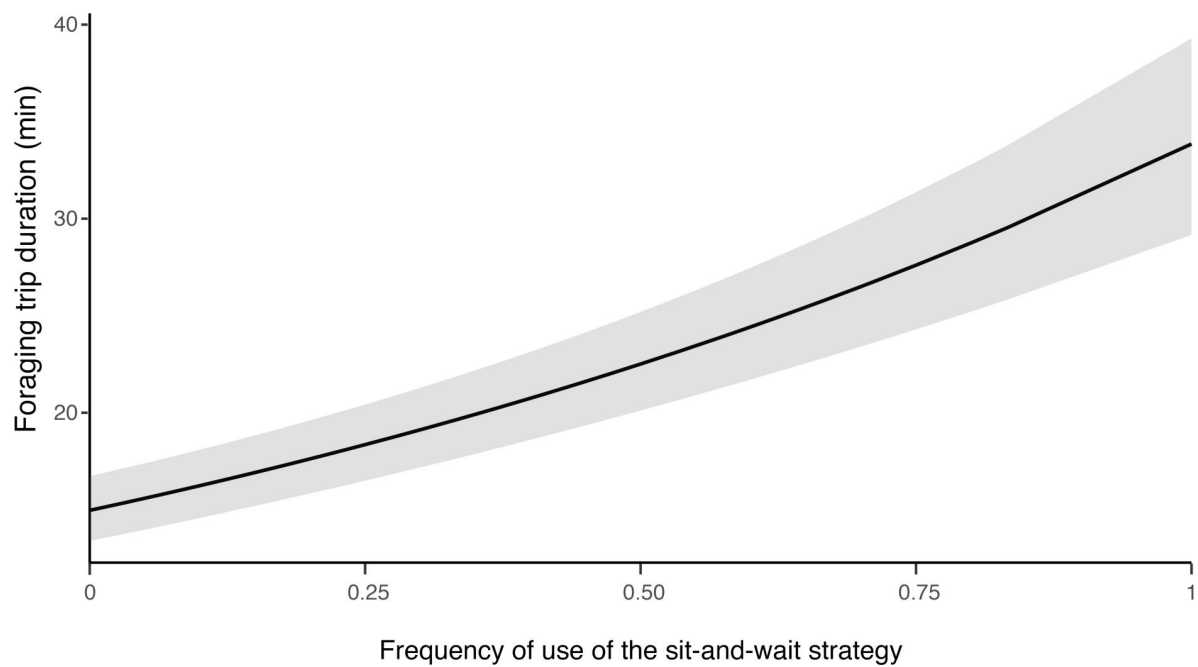


Figure S11

Influence of hunting strategy on barn owls foraging trip duration.

Variation in foraging trip duration (in min) as a function of the frequency of use of the sit-and-wait strategy (relative to hunting on the wing). Solid line shows the estimated mean (averaged over both sexes) and the grey shade corresponds to the 95% confidence intervals around the mean.

Flight	BE	Present	Range	Flexibility	ETNE
TS1	Take-off	25	26	50	
TS2	Flying	5	6	60'000	10
TS3	Landing	90	91	5	
TS4	Standing	4			
Hunting strike	BE	Present	Range	Flexibility	ETNE
TS1	Tilt head down	5	20	40	10
TS2	Legs swinging	3	5	110	
TS3	Impact	1	1	160	
TS4	Standing	5			
Nest return	BE	Present	Range	Flexibility	ETNE
TS1	Nest enter	1	3	4	
TS2	Nest in	1	50	1500	5
TS3	Nest exit	1			
Self feeding	BE	Present	Range	Flexibility	ETNE
TS1	High peak	1	1	15	
TS2	Low peak	1	1	15	
TS3	High peak	1	1	15	
TS4	Low peak	1	1	15	
TS5	Low VeDBA	10	10	150	
TS6	End	15			

Table S1.

Flexible time algorithms in a Boolean approach for classification of barn owl behaviours.

Time series (TS) of base elements applied for the classification of flight, hunting strikes, nest return (indicating prey delivery to the nest), and self-feeding behaviours. Each base element (BE) has a defined temporal flexibility that includes a number of events over which the conditions of the element are met (Present), as well as a range to the next element (Range), a window of flexibility (Flexibility) and a period over which the element is extended (ETNE), where the units are in events, 50 Hz having 50 events per second. See Wilson et al.⁽⁵⁴⁾ for details of use of these algorithms in the Boolean approach.

Table S2

Modelling the effect of landing context and sex on landing force.

Model output for the LMM predicting variations in landing force (log transformed) due to landing context (hunting strike vs perching) and sex (Males vs Females). The model also included a random effect of BirdID and NightID (nested in BirdID). Intercept is reported for both landing contexts (highlighted in grey) and give information about the averaged landing force considering female individuals. Estimates for interactions give the % of change between females and males for each landing contexts. Variance (σ^2), intra-class correlation coefficient (ICC) and number of observations are provided for random effects.

<i>Predictors</i>	impact force (N)		
	<i>Estimate</i>	<i>Conf. Int.</i>	<i>p-value</i>
landing context [strike]	40.81	39.49 – 42.18	<0.001
landing context [perching]	9.94	9.63 – 10.27	<0.001
landing context [strike] * sex[M]	0.94	0.90 – 0.99	0.011
landing context [perching] * sex[M]	0.80	0.76 – 0.83	<0.001
Observations	84855		
Marginal R ² / Conditional R ²	0.736 / 0.772		
Random Effects: $\sigma^2 = 0.15$ ICC = 0.14 N _{BirdID} = 163 N _{NightID} = 7			

Table S2b

Model selection results using Akaike Information Criteria corrected for small sample sizes (AICc) for all possible models evaluating the effect of landing context and sex on barn owl landing force.

A “*” in the variable columns indicates that the variable was included in that model. K is the number of variables included in each model. All LMM included a random effect of BirdID and NightID (nested in BirdID). Table includes all possible models, ranked by AICc. Models with Delta AICc < 2 are highlighted in grey and the final model is shown in bold.

#	Sex	Landing context	Sex : Landing context	K	AIC	ΔAIC	Model weight
1	*	*	*	3	82517	0	1
2	*	*		2	83252.5	735.51	0
3	*			1	83297.9	780.84	0
4		*		1	198495.8	115978.81	0
5				0	198498.4	115981.36	0

LME			
impact force (N)			
<i>Predictors</i>	<i>Estimate</i>	<i>Conf. Int.</i>	<i>p-value</i>
intercept	8.99	8.736 – 9.242	<0.001
perch type [tree]	1.07	1.068 – 1.080	<0.001
perch type [buildings]	1.09	1.079 – 1.094	<0.001
wind speed	1.00	0.99 – 1.00	0.04
sex [M]	0.85	0.81 – 0.88	<0.001
GAM			
<i>Predictors</i>	<i>EDF</i>	<i>F</i>	<i>p-value</i>
Time to hunt: perch type [road/pasture]	4.22	22.43	<0.001
Time to hunt: perch type [tree]	1.50	5.12	0.005
Time to hunt: perch type [buildings]	1.00	12.86	<0.001
Observations	40305		
R ²	0.235		

Table S3a

Modelling the effect of time to hunt and perch type on landing force during perching events.

Model output of the GAMM predicting variations in pre-hunt perching force in N (log transformed) due to perch type (buildings, tree branches and road/pasture poles), wind speed, and sex as linear predictor (lme) and time to the next strike as an additive term (gam). The model also included a random effect of BirdID. Effective degrees of freedom (EDF) are shown for additive terms, providing the degree of non-linearity between pre-hunt perching force and time to hunt for each perch type.

#	Perch type	Wind speed	Sex	s(time to hunt : perch type)	K	AICc	ΔAICc	Model weight
1	*	*	*	*	4	203.1	0	0.48
2	*	*		*	3	204.5	1.42	0.24
3	*		*	*	3	205.0	1.92	0.18
4	*			*	2	206.4	3.26	0.09
5	*	*	*		3	324.2	121.13	0

Table S3b

Model selection results using Akaike Information Criteria corrected for small sample sizes (AICc) for all possible models evaluating the effect of time to hunt and perch type on landing force during perching events.

A “*” in the variable columns indicates that the variable was included in that model. K is the number of variables included in each model. All LMM included a random effect of BirdID and NightID (nested in BirdID). Table includes the top five models, ranked by AICc. Models with Delta AICc < 2 are highlighted in grey and the final model is shown in bold.

Table S4a

Modelling the effect of hunting success and hunting strategy on landing force during hunting strike.

Model output for the LMM predicting variations in hunting strike force (log transformed) due to hunting success (0 vs 1), hunting strategy (perching vs flying) and sex. The model also included a random effect of BirdID and NightID (nested in BirdID). Intercept provides the averaged strike force (N) considering female individuals hunting on the wing. Variance (σ^2), intra-class correlation coefficient (ICC) and number of observations are provided for random effects.

<i>Predictors</i>	impact force (N)		
	<i>Estimate</i>	<i>Conf. Int.</i>	<i>p-value</i>
Intercept	41.51	40.31 – 42.75	<0.001
hunting success [0]	0.95	0.94 – 0.96	<0.001
sex [M]	0.94	0.91 – 0.98	0.005
hunting strategy [sit-and-wait]	0.96	0.94 – 0.99	0.002
hunting success [0] *	1.04	1.01 – 1.07	0.004
hunting strategy [sit-and-wait]			
Observations	27981		
Marginal R ² / Conditional R ²	0.007 / 0.115		
Random Effects: $\sigma^2 = 0.13$ ICC = 0.11 N _{BirdID} = 163 N _{NightID} = 7			

Table S4b

Model selection results using Akaike Information Criteria corrected for small sample sizes (AICc) for all possible models evaluating the effect of hunting success and hunting strategy on landing force during hunting strike.

A “*” in the variable columns indicates that the variable was included in that model. K is the number of variables included in each model. All LMM included a random effect of BirdID and NightID (nested in BirdID). Table includes the top five models, ranked by AICc. Models with Delta AICc < 2 are highlighted in grey and the final model is shown in bold.

#	Sex	Hunting success	Hunting strategy	Hunting strategy: Hunting success	K	AIC	Δ AIC	Model weight
1	*	*	*	*	4	23996.3	0	0.87
2	*	*			2	24002.1	5.82	0.05
3		*	*	*	3	24002.1	5.82	0.05
4	*	*	*		3	24002.7	6.36	0.04
5		*			1	24007.7	11.41	0.003

Table S5a

Modelling the effect of sex and hunting strategy on hunting success.

Model output for the GLMM predicting variations in hunting success (binary response 0,1) due to sex and hunting strategy (on the wing vs sit-and-wait). The model also included a random effect of BirdID and NightID (nested in BirdID). Intercept is reported for both sexes and give information about the averaged hunting success considering individuals hunting on the wing. Standardized estimates are provided for any additional terms in the model, representing % of change (odds ratio) of hunting success. The influence of landing force on hunting success is provided considering both hunting strategies. Variance (σ^2), intra-class correlation coefficient (ICC) and number of observations of each group are provided for random effects.

<i>Predictors</i>	Hunting success		
	<i>Estimate</i>	<i>Conf. Int.</i>	<i>p-value</i>
sex [Female]	0.24	0.22 – 0.26	<0.001
sex [Male]	0.35	0.33 – 0.38	<0.001
Hunting strategy [sit-and-wait]: Females	1.55	1.38 – 1.75	<0.001
Hunting strategy [sit-and-wait]: Males	1.49	1.32 – 1.68	<0.001
Observations	27981		
Marginal R ² / Conditional R ²	0.014 / 0.051		
Random effects: $\sigma^2 = 3.29$ ICC = 0.04 N _{BirdID} = 163 N _{NightID} = 7			

Table S5b

Model selection results using Akaike Information Criteria corrected for small sample sizes (AICc) for all possible models evaluating the effect of sex, hunting strategy, and their interaction on hunting success.

A “*” in the variable columns indicates that the variable was included in that model. K is the number of variables included in each model. All LMM included a random effect of BirdID and NightID (nested in BirdID). Table includes the top five models, ranked by AICc. Models with Delta AICc < 2 are highlighted in grey and the final model is shown in bold.

#	Sex	Hunting strategy	Sex : Hunting strategy	K	AIC	Δ AIC	Model weight
1	*	*		2	30726.8	0	0.71
2	*	*	*	3	30728.6	1.79	0.29
3	*			1	30768.0	41.22	0
4		*		1	30816.2	89.38	0
5				0	30848.5	121.68	0

<i>Predictors</i>	Hunting success		
	<i>Estimate</i>	<i>Conf. Int.</i>	<i>p-value</i>
Females	0.22	0.18 – 0.26	<0.001
Males	0.32	0.28 – 0.37	<0.001
hunting strategy [sit-and-wait]	1.51	1.26 – 1.81	<0.001
hunting strategy [on the wing]: pre-hunt perching force	1.08	0.97 – 1.20	0.178
hunting strategy [sit-and-wait]: pre-hunt perching force	0.85	0.73 – 0.99	0.037
Observations	3040		
Marginal R ² / Conditional R ²	0.023 / 0.043		
Random Effects: $\sigma^2 = 3.29$ ICC = 0.02 N _{BirdID} = 151 N _{NightID} = 7			

Table S6a

Modelling the effect of landing force during pre-hunt perching on hunting success.

Model output for the GLMM predicting variations in hunting success (binary response 0,1) due to pre-hunt perching force, sex, hunting strategy (on the wing vs sit-and-wait) and wind speed. The model also included a random effect of BirdID and NightID (nested in BirdID). Intercept is reported for both sexes (highlighted in grey) and give information about the averaged hunting success considering individuals hunting on the wing. Standardized estimates are provided for any additional terms in the model, representing % of change (odds ratio) of hunting success. The influence of landing force on hunting success is provided considering both hunting strategies. Variance (σ^2), intra-class correlation coefficient (ICC) and number of observations of each group are provided for random effects.

Table S6b

Model selection results using Akaike Information Criteria corrected for small sample sizes (AICc) for all possible models evaluating the effect of landing force during pre-hunt perching on hunting success.

A “*” in the variable columns indicates that the variable was included in that model. K is the number of variables included in each model. All LMM included a random effect of BirdID and NightID (nested in BirdID). Table includes the top five models, ranked by AICc. Models with Delta AICc < 2 are highlighted in grey and the final model is shown in bold.

#	Sex	Wind speed	Perch type	Hunting strategy	Pre-hunt perching force	Hunting strategy : Pre-hunt perching force	K	AIC	ΔAIC	Model weight
1	*			*	*	*	4	3326.5	0	0.24
2	*		*	*	*	*	5	3327.1	0.54	0.19
3	*	*		*	*	*	5	3327.4	0.82	0.16
4	*	*	*	*	*	*	6	3328.1	1.55	0.11
5	*			*			2	3329.1	2.56	0.07

Table S7.

Modelling the effect of sex on foraging flights speed.

Model output of the LMM predicting variation of barn owls foraging flights speed (in ms^{-1}) as a function of the sex. Intercept provides the averaged foraging flight speed (in ms^{-1}) considering female individuals. The model also included a random effect of BirdID and NightID (nested in BirdID). Variance (σ^2), intra-class correlation coefficient (ICC) and number of observations of each group are provided for random effects.

<i>Predictors</i>	Foraging flight speed (ms^{-1})		
	<i>Estimate</i>	<i>Conf. Int.</i>	<i>p-val</i>
Intercepts	5.47	5.38 – 5.56	<0.001
Sex [Male]	-0.23	-0.36 – -0.10	<0.001
Observations	27242		
Marginal R ² / Conditional R ²	0.006 / 0.141		
Random Effects: $\sigma^2 = 1.60$ ICC = 0.14 $N_{\text{BirdID}} = 163$ $N_{\text{NightID}} = 7$			

<i>Predictors</i>	Foraging trip duration (ms ⁻¹)		
	<i>Estimate</i>	<i>Conf. Int.</i>	<i>p-value</i>
(Intercept)	9.49	8.43 – 10.67	<0.001
nb of hunting attempts	1.16	1.16 – 1.17	<0.001
freq sit-and-wait	2.26	1.97 – 2.59	<0.001
sex [Male]	0.79	0.69 – 0.90	0.001
Observations	3915		
Marginal R ² / Conditional R ²	0.299 / 0.402		
Random effects: $\sigma^2 = 1$ ICC = 0.15 N _{birdID} = 150 N _{NightID} = 7			

Table S8

Modelling the effect of sex and hunting strategy on barn owls foraging trip duration.

Model output of the LMM predicting variation of barn owls foraging trip duration (in min) as a function of the total number of hunting attempts per trip, the frequency of use of the sit-and-wait strategy (relative to hunting on the wing), and sex. The model also included a random effect of BirdID and NightID (nested in BirdID). Variance (σ^2), intra-class correlation coefficient (ICC) and number of observations of each group are provided for random effects.

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Reviewer #1 (Public Review):

In this paper, Schalcher et al. examined how barn owls' landing force affects their hunting success during two hunting strategies: strike hunting and sit-and-wait hunting. They tracked tens of barn owls that raised their nestlings in nest boxes and utilized high-resolution GPS and acceleration loggers to monitor their movement. In addition, camcorders were placed near their nest boxes and used to record the prey they brought to the nest, thus measuring their foraging success.

This study generated a unique dataset and provided new insights into the foraging behavior of barn owls. The researchers discovered that the landing force during hunting strikes was significantly higher compared to the sit-and-wait strategy. Additionally, they found a positive relationship between landing force and foraging success during hunting strikes, whereas, during the sit-and-wait strategy, there was a negative relationship between the two. This suggests that barn owls avoid detection by generating a lower landing force and producing less noise. Furthermore, the researchers observed that environmental characteristics affect barn owls' landing force during sit-and-wait hunting. They found a greater landing force when landing on buildings, a lower landing force when landing on trees, and the lowest landing force when landing on poles. The landing force also decreased as the time to the next hunting attempt decreased. These findings collectively suggest that barn owls reduce their landing force as an acoustic camouflage to avoid detection by their prey.

The main strength of this work is the researchers' comprehensive approach, examining different aspects of foraging behavior, including high-resolution movement, foraging success, and the influence of the environment on this behavior, supported by impressive data collection.

The results presented support the authors' conclusion that lower landing force during sit-and-wait hunting increases hunting success, likely due to a decreased probability of detection by

their prey, resulting in acoustic camouflage. The authors also hypothesized that hunting success is crucial for survival, and thus, acoustic camouflage has a direct link to fitness. This paper provides an unprecedented dataset and the first measurement of landing force during hunting in the wild. It is likely to inspire many other researchers currently studying animal foraging behavior to explore how animals' movement affects foraging success.

<https://doi.org/10.7554/eLife.87775.2.sa1>

We would like to thank you and the reviewers for your thoughtful comments that assisted us to improve the manuscript. We carefully followed the reviewers' recommendations and provide a detailed point-by-point account of our responses to the comments.

Please find below the important changes in the updated manuscript.

- (1) We changed the title according to the comments provided by reviewer #1.
- (2) We edited the introduction, results, and discussion to improve the link between the objectives of the study, the findings, and their discussion, as reviewer #2 recommended.
- (3) We clarified the link between camouflage and fitness, which is now presented as a hypothesis, as reviewer #1 suggested.
- (4) We added new analyses and figures in the main text and in the supplementary materials to better emphasize sex differences in landing force, foraging strategies and hunting success, following reviewer #1 suggestion.
- (5) According to reviewer #2 comments, we edited the results adding key information about methods to help the reader understand the findings without reading the Methods section.
- (6) We added important details about the model selection approach along with a discussion of the low R-square values reported in our analyses on hunting success, as reviewer #2 suggested.

eLife assessment

This fundamental work substantially advances our understanding of animals' foraging behaviour, by monitoring the movement and body posture of barn owls in high resolution, in addition to assessing their foraging success. With a large dataset, the evidence supporting the main conclusions is convincing. This work provides new evidence for motion-induced sound camouflage and has broad implications for understanding predator-prey interactions.

Public Reviews:

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The main strength of this work is the researchers' comprehensive approach, examining different aspects of foraging behavior, including high-resolution movement, foraging success, and the influence of the environment on this behavior, supported by impressive data collection. The weakness of this study is that the results only present a partial biological story contained within the data. The focus is on acoustic camouflage without addressing other aspects of barn owls' foraging strategy, leaving the reader with many unanswered questions. These include individual differences, direct measurements of owls' fitness, a detailed analysis of the foraging strategy of males and females, and the collective effort per nest box. However, it is possible that these data will be published in a separate paper.

We greatly appreciate your recognition of the comprehensive approach and extensive data collection. Our primary objective was to study the role of acoustic camouflage. Nonetheless, the manuscript now includes a detailed analysis of the foraging strategy and hunting success of males and females (lines 164-225).

The results presented support the authors' conclusion that lower landing force during sit-and-wait hunting increases hunting success, likely due to a decreased probability of detection by their prey, resulting in acoustic camouflage. The authors also argue that hunting success is crucial for survival, and thus, acoustic camouflage has a direct link to fitness. While this statement is reasonable, it should be presented as a hypothesis, as no direct evidence has been provided here.

Thank you for the comment. We agree and thus have edited the language accordingly.

However, since information about nestling survival is typically monitored when studying behavior during the breeding period, the authors' knowledge of the effect of acoustic camouflage on owls' fitness can probably be provided. Furthermore, it will be interesting to further examine the foraging strategies used by different individuals during foraging, the joint foraging success of both males and females within each nest box, and the link between landing force and foraging success if the data are available.

We are currently writing a manuscript on these topics. We are aware that several scientific questions regarding the foraging ecology of the barn owl still need our attention. Regarding the link between landing force and foraging success, we believe that our revised manuscript addresses this specific topic, please see specific responses below.

However, even without this additional analysis on survival, this paper provides an unprecedented dataset and the first measurement of landing force during hunting in the wild. It is likely to inspire many other researchers currently studying animal foraging behavior to explore how animals' movements affect foraging success.

Reviewer #2 (Public Review):

Summary:

The authors provide new evidence for motion-induced sound camouflage and can link the hunting approach to hunting success (detailing the adaptation and inferring a fitness consequence).

Strengths:

Strong evidence by combining high-resolution accelerometer data with a ground-truthed data set on prey provisioning at nest boxes. A good set of co-variables to control for some of the noise in the data provides some additional insights into owl hunting attempts.

Weaknesses:

There is a disconnect between the hypotheses tested and the results presented, and insufficient detail is provided on the statistical approach. R2 values of the presented models are very small compared to the significance of the effect presented. Without more detail, it is impossible to assess the strength of the evidence.

In the revised manuscript, we changed the way results are presented and we improved the link between the hypotheses and the results. The R2 values are indeed small. It is however important to keep in mind that we are assessing the outcome of one specific behavior (i.e. landing force during sit-and-wait hunts) on hunting success in a wild environment, where many complex ecological interactions likely influence hunting success. Nonetheless, the coefficients (as reported in the results) show that for every 1 N increase in landing force, there is a 15% reduction in hunting success, which is substantial. In the discussion we also note that 50 Hz is a relatively low sampling frequency for estimating the peak ground reaction force. We have gone back over the presentation of our results and made our discussion more nuanced to acknowledge this aspect.

We have also added a detailed description about our model selection process in the methods section and provide a model selection table for each analysis in the supplementary materials.

The authors seem to overcome persisting challenges associated with the validation and calibration of accelerometer data by ground-truthing on-board measures with direct observations in captivity, but here the methods are not described any further and sample sizes (2 owls - how many different loggers were deployed?) might be too small to achieve robust behavioural classifications.

Thank you for the comment. Details of our methods of behavioural identification are provided in lines 385 – 429. There are two reasons why our results should not be limited by the sample size. First, we used the temporal sequence of changes in acceleration, and rates of change in acceleration data, which make the methods robust to individual differences in acceleration values. Furthermore, our methods for behavioural identification were not based on machine learning. Instead, we use a Boolean based approach (as described in Wilson et al. 2018. MEE), which is more robust to small differences in absolute values that might occur e.g. in relation to slight changes in device position.

Recommendation for the authors:

Reviewer #1 (Recommendations For The Authors):

Comment 1. This study provides new insights into animals' foraging behavior and will probably inspire other researchers to examine foraging behavior in such high resolution.

We hope so, thank you.

Comment 2. However, it is necessary to describe better the measured landing force and the hunting strike and perching behavior so the readers can understand these methods when reading the results (and without reading the Methods).

We have now changed the text in the “Results” to help the reader understand the key methods while reading the results.

Comment 3. In addition, make sure you use the same terminology for hunting strategies during the entire paper and especially in all figures and corresponding result descriptions.

We now use consistent terminology throughout the text and figures. We hope that this is now clear in the revised manuscript.

Comment 4. In addition, although I find your statement about the link between acoustic camouflage and fitness reasonable, it should be described as a hypothesis or examined if you want to keep the direct link statement. I believe showing a direct link can add an additional outstanding aspect to this paper, but I also understand that it can be addressed in a separate paper.

We agree that the relationship between hunting success and barn owl fitness is an important topic, but it necessitates a consideration of both hunting strategies, including hunting on the wing, which extends beyond the limits of our current study. Indeed, our primary objective was to conduct a detailed examination of the interplay between acoustic camouflage and the success of the sit-and-wait technique.

However, we have edited the manuscript to explicitly describe the link between acoustic camouflage and fitness as a hypothesis. We believe this adjustment provides a more accurate representation of our approach. We hope this clarifies the specific emphasis of our work and its contribution to the understanding of barn owl hunting behavior.

Here are my detailed comments about the paper:

Comment 5. Title: Consider changing the title to "Acoustic camouflage predicts hunting success in a wild predator."

We would like to thank you for your nice proposition. However, we opted for a different title, which is now “Landing force reveals new form of motion-induced sound camouflage in a wild predator”.

Comment 6. Line 91-93: Please provide additional information about the collected dataset, including:

Description of the total period of observations, an average and standard deviation of perching and hunting attempt events per individual per night, number of foraging trips per individual per night, details about the geographic location and characteristics of the habitat, season, and reproductive state.

The revised manuscript now includes detailed information about the collected dataset (i.e. study area, reproductive state, etc...). “We used GPS loggers and accelerometers to record high resolution movement data during two consecutive breeding seasons (May to August in 2019 and 2020) from 163 wild barn owls (79 males and 84 females) breeding in nest boxes across a 1,000 km² intensive agricultural landscape in the western Swiss plateau.” Results section, lines 79 – 82

Details about the number of foraging trips per individuals and per night are now presented in the results: “Sexual dimorphism in body mass was marked among our sampled individuals. Males were lighter than females (84 females, average body mass: 322 ± 22.6 g; 79 males, average body mass 281 ± 16.5 g, Fig S6) and provided almost three times more prey per night than females (males: 8 ± 5 prey per night; females: 3 ± 3 prey per night; Fig.S7). Males also displayed higher nightly hunting effort than females (Males: 46 ± 16 hunting attempts per night, $n = 79$; Females: 25 ± 11 hunting attempts per nights, $n = 84$; Fig. 3A, Fig S8). However, females were more likely to use a sit and wait strategy than males (females: $24\% \pm 15\%$, males: $13\% \pm 10\%$, Fig.S9). As a result, the number of perching events per night was similar between males and females (Females: 76 ± 23 perching events per nights; Males: 69 ± 20 perching events per night; Fig S8).” (lines 165 – 174)

Comment 7. In addition, state if the information describes breeding pairs of males and females and provides statistics on the number of tracked pairs and the number of nest boxes.

The revised manuscript now includes a description of the number of tracked breeding pairs and the number of nest boxes. “Of these individuals, 142 belonged to pairs for which data were recovered from both partners (71 pairs in total, 40 in 2019, 31 in 2020). The remaining 21 individuals belonged to pairs with data from one partner (11 females and 1 male in 2019; 4 females and 5 males in 2020).” (lines 82 – 85.)

Comment 8. Line 93: Briefly define the term "landing force" and explain how it was measured (and let the reader know that there is a detailed description in the Methods).

We now include a brief definition of the “landing force” along with a brief explanation of how it was measured in the results section. “We extracted the peak vectoral sum of the raw acceleration during each landing and converted this to ground reaction force (hereafter “landing force”, in Newtons) using measurements of individual body mass (see methods for detailed description).” (lines 92 – 95).

Comment 9. Line 94: All definitions, including "pre-hunting force," need to be better described in the Results section.

Thank you for this suggestion. We now provided a better description of those key definitions directly in the results section:

Measurement of landing force: “Barn owls employing a sit-and-wait strategy land on multiple perches before initiating an attack, with successive landings reducing the distance to the target prey (Fig. 2C).

We used the acceleration data to identify 84,855 landings. These were further categorized into perching events ($n = 56,874$) and hunting strikes ($n = 27,981$), depending whether barn owls were landing on a perch or attempting to strike prey on the ground (Fig. 1A and B, see methods for specific details on behavioral classification).” (lines 88 – 95)

Pre-hunt perching force predicts hunting success: “Finally, we analyzed whether the landing force in the last perching event before each hunting attempt (i.e. pre-hunt perching force) predicted variation in hunting success” (lines 229 – 230)

Comment 10. Line 102: Remove "Our analysis of 27,981 hunting strikes showed that" and add "n = 27,981" after the statistics. You have already stated your sample size earlier. There is no need to emphasize it again, although your sample size is impressive.

We modified the text in the results section as suggested.

Comment 11. Line 104: The results so far suggest that the difference in landing force between males and females is an outcome of their different body masses. However, it is not clear what is the reason for the difference in the number of hunting strike attempts between males and females (Lines 104-106). Can you compare the difference in landing force between males and females with similar body mass (females from the lower part of the distribution and males from the upper part)? Is there still a difference?

Thank you, following your comment we made some new analyses that clarified the situation around landing force involved in perching and hunting strike events between sexes. But firstly, we wanted to clarify why there is a difference in number of hunting attempts between males and females. During the breeding season, females typically perform most of the incubation, brooding, and feeding of nestlings in the nest, while the male primarily hunts food for the female and chicks. The female supports the male providing food in a very irregular way, and this changes from pair to pair (paper in prep.). The differences in number of hunting attempts between males and females reflects this asymmetry in food provisioning between sexes during this specific period. We specified this in the revised version of the manuscript (lines 164 – 174).

We also provide a new analysis to investigate sex differences in mass-specific landing force (force/body mass). We found that males and females produce similar force per unit of body mass during perching events. This demonstrates that the overall higher perching force in females (see Fig. 4C in the manuscript) is therefore driven by their higher body mass. (lines 194 – 199)

Comment 12. Line 154: I believe Boonman et al. (2018) is relevant to this part of the discussion. Boonman, Arjan, et al. found that barn owl noise during landing and taking off is worth considering. ["The sounds of silence: barn owl noise in landing and taking off."]

Behavioral Processes 157 (2018): 484-488.]

We now cited this paper in the discussion.

Comment 13. Line 164: Your results do not directly demonstrate a link to fitness, although they potentially serve as a proxy for fitness (add a reference). However, you might have information regarding nestlings' survival - that will provide a direct link for fitness. Change your statement or add the relevant data.

We appreciated your feedback, and we adjusted the language accordingly.

Comment 14. Line 213: If the poles are closer to the ground - is it possible that the higher trees and buildings serve for resting and gathering environmental information over greater distances? For example, identifying prey at farther distances or navigating to the next pole?

Yes, this is indeed the most likely explanation for the fact that owls land more on buildings and trees than on poles until the last period (about 6 minutes) before hunting. In these last minutes, barn owls preferentially use poles, as we showed in figure 2B. The revised manuscript now includes this explanation in the discussion (lines 269 – 284).

Comment 15. Line 250: The product "AXY-Trek loggers" does not appear on the Technosmart website (there are similar names, but not an exact match). Are you sure this

is the correct name of the tracking device you used?

Thank you for pointing out this detail that we missed. The device we used is now called "AXY-Trek Mini" (<https://www.technosmart.eu/axy-trek-mini/>). We have corrected this error directly in the revised manuscript.

Comment 16. Line 256: Please explain how the devices were recovered. Did you recapture the animals? If so, how? Additionally, replace "after approximately 15 days" with the exact average and standard deviation. Furthermore, since you have these data, please state the difference in body mass between the two measurements before and after tagging.

The birds were recaptured to recover the devices. Adults barn owls were recaptured at their nest sites, again using automatic sliding traps that are activated when birds enter the nest box. The statement "after approximately 15 days" was replaced by the exact mean and standard deviation, which were 10.47 ± 2.27 days. Those numbers exclude five individuals from the total of 163 individuals included in this study. They could not be recaptured in the appropriate time window but were re-encountered when they initiated a second clutch later in the season (4 individuals) or a new clutch the year after (1 individual).

We integrated this previously missing information in the revised manuscript (lines 370 – 372).

Comment 17. Line 259: What was the resolution of the camera? What were the recording methods and schedule? How did you analyze these data?

The resolution was set to 3.1 megapixel. Motion sensitive camera traps were installed at the entrance to each nest box throughout the period when the barn owls were wearing data loggers, and each movement detected triggered the capture of three photos in bursts. The photos recorded were not analyzed as such for this study, but were used to confirm each supply of prey, which had previously been detected from the accelerometer data. We added these details in the revised manuscript (lines 377 – 380)

Comment 18_1. Figure 1:

Panel A) Include the sex of the described individual.

The sex of the described individual is now included in the figure caption.

Comment 18_2. It would be interesting to show these data for both males and females from the same nest box (choose another example if you don't have the data for this specific nest box).

Although we agree that showing tracks of males and females from the same nest is very interesting, the purpose of this figure was to illustrate our data annotation process and we believe that adding too many details on this figure will make it appear messy. However, the revised manuscript now includes a new figure (Fig. 3A) which shows simultaneous GPS tracks of a male and a female during a complete night, with detailed information about perching and hunting behaviors.

Comment 18_3. Add the symbol of the nest box to the legend.

Done

Comment 18_4. Provide information about the total time of the foraging trip in the text below.

The duration of the illustrated foraging trip has been included in the figure caption.

Comment 18_5. To enhance the figure's information on foraging behavior, consider color coding the trajectory based on time and adding a background representing the landscape. Since this paper may be of interest to researchers unfamiliar with barn owl foraging behavior, it could answer some common questions.

For similar reasons explained in our answer above (Comment 18_2), we would rather keep this figure as clean as possible. However, we followed your recommendations and included these details in the new Figure 3 described above. In this new figure, GPS tracks are color coded according to the foraging trip number and includes a background representing the landscape. To provide even more detail about the landscape, we added another figure in the supplementary materials (Fig. S2) which provides illustration of barn owls foraging ground and nest site that we think might be of interest for people unfamiliar with barn owls.

Comment 18_6. Inset panels) provide a detailed description of the acceleration insert panels.

Done

Comment 18_7. Color code the acceleration data with different colors for each axis, add x and y axes with labels, and ensure the time frame on the x-axis is clear. How was the self-feeding behavior verified (should be described in the methods section)?

We kept both inset panels as simple as possible since they serve here as examples, but a complete representation of these behaviors (with time frame, different colors and labels) is provided in the supplementary materials (figure S3). We included this statement in the figure caption and added a reference to the full representations from the supplementary materials:

In the Figure caption: "Inset panels show an example of the pattern of the tri-axial acceleration corresponding to both nest-box return and self-feeding behaviors (but see Fig S3 for a detailed representation of the acceleration pattern corresponding to each behavior)."

In the Method section: "Self-feeding was evident from multiple and regular acceleration peaks in the surge and heave axes (resulting in peaks in VeDBA values > 0.2 g and < 0.9 g, Fig.S3D), with each peak corresponding to the movement of the head as the prey was swallowed whole."

Comment 18_8. Panel B) Note in the caption that you refer to the acceleration z-axis.

We believe that keeping the statement "the heave acceleration..." in the figure caption is more informative than referring to the "z-axis" as it describes the real dimension to which we are referring. The use of the x, y and z axes can be misleading as they can be interchanged depending on the type and setting of recorders used.

Comment 18_9. Present the same time scale for both hunting strategies to facilitate comparison. You can achieve this by showing only part of the flight phase before perching.

Done

Comment 18_10. Panel C) Presenting the data for both hunting strategy and sex would provide more comprehensive information about the results and would be relatively easy to implement.

We agree with your comment. We present the differences in landing force for both landing contexts and sexes in the new Figure 3 as well as in the supplementary materials (Figure S10) of this revised manuscript.

Comment 19. Figure 2: Please provide an explanation of the meaning of the circles in the figure caption.

Done

Comment 20. Figure 3:

Panel A) It is unclear how the owl illustration is relevant to this specific figure, unlike the previous figures where it is clear. Also, suggest removing the upper black line from the edge of the figure or add a line on the right side.

Done (now in Figure 2).

Panel B) "Density" should be capitalized.

Done

Panel C) Add a scale in meters, and it would be helpful to include an indication of time before hunting for each data point.

Done

Comment 21. Figure S1: Mark the locations of the nest boxes and ensure that trajectories of different individuals and sexes can be identified.

The purpose of this figure was to show the spatial distribution of the data. We think that adding nest locations and coloring the paths according to individuals and/or sex will make the figure less clear. However, the new Figure 3 highlights those details.

Comment 22. Figure S2: Show the pitch angle similarly to how you showed the acceleration axes, and explain what "VeDBA" stands for. Provide a description of the perching behavior, clearly indicating it on the figure. Add axes (x, y, z) to the illustration of the acceleration explanation.

We edited this figure (now figure S3) to show the pitch angle and provide an explanation of what "VeDBA" stands for in the figure caption. The figure caption now also provides a better description of the perching behavior. For the axes (i.e. X, Y, Z), we prefer to refer to the heave, surge, and sway as this is more informative and refers to what is usually reported in studies working with tri-axial accelerometers.

Comment 23. Table S1: Improve the explanation in the caption and titles of the table.

Done

Reviewer #2 (Recommendations For The Authors):

Comment 1. From the public review and my assessment there, the authors can be assured that I thoroughly enjoyed the read and am looking forward to seeing a revised and improved version of this paper.

We thank the reviewer for this comment. We revised the manuscript according to their comments.

Comment 2. In addition to my major points stated above, I would like to add the following recommendations:

The manuscript is overall well written, but it uses a very pictorial language (a little as if we were in a David Attenborough documentary) that I find inappropriate for a research paper (especially in the abstract and introduction, "remarkable" (2x), "sophisticated" (are there any unsophisticated adaptations? We are referring to something under selection after all) etc.

We appreciated that you found the paper overall well written, and we understand the comment about pictorial language. We therefore slightly changed the text to make sure that the adjective used to describe adaptive strategies are not over-emphasized.

Comment 3. Abstract

"While the theoretical benefits of predator camouflage are well established, no study has yet been able to quantify its consequences for hunting success." - This claim is actually not fully true:

Nebel Carina, Sumasgutner Petra, Pajot Adrien and Amar Arjun 2019: Response time of an avian prey to a simulated hawk attack is slower in darker conditions, but is independent of hawk colour morph. Soc. open sci.6:190677

We edited our claim to specify that the consequences of predator camouflage on hunting success has never been quantified in natural conditions and cited the reference in the introduction.

Comment 4. Line 23. Rephrase to: "We used high-resolution movement data to quantify how barn owls (Tyto alba) conceal their approach when using a sit-and-wait strategy, as well as the power exerted during strikes."

We edited this sentence in the abstract, as suggested.

Comment 5. Results

There is a disconnect between the objectives outlined at the end of the introduction and the following results that should be improved.

The authors state: "Using high-frequency GPS and accelerometer data from wild barn owls (Tyto alba), we quantify the landing dynamics of this sit-and-wait strategy to (i) examine how birds adjust their landing force with the behavioral and environmental context and (ii) test the extent to which the magnitude of the predator cue affects hunting success." But one of the first results presented are sex differences.

This is a fair point. We have now changed our statement in the end of the introduction as well as the order of the results to improve the link between the objectives outlined in the introduction and the way result are presented.

Comment 6. At this stage, the reader does not even know yet that we are presented with a size-dimorphic species that also has very different parental roles during the breeding season. This should be better streamlined, with an extra paragraph in the introduction. And these sex differences are then not even discussed, so why bring them up in the first

place (and not just state "sex has been fitted as additional co-variate to account for the size-dimorphism in the species" without further details).

We edited the way the objectives are outlined in the introduction to cover the size dimorphism (lines 70 – 76). We also completely changed the way the sex differences are presented in the results, including a new analysis that we believe provides a better comprehensive understanding of barn owl foraging behavior (lines 164 – 206). Finally, we added a new paragraph in the discussion to consider those results (lines 319 – 339).

Comment 7. It is not clear to me where and how high-resolution GPS data were used? The results seem to concentrate on ACC – why GPS was used and how it features should be foreshadowed in a few lines in the introduction. I definitively prefer having the methods at the end of a manuscript, but with this structure, it is crucial to give the reader some help to understand the storyline.

GPS data were used to validate some behavioral classifications (prey provisioning for example), but most importantly they were used to link each landing event with perch types. We edited the text in the result section to clarify where GPS and/or ACC data were used.

Comment 8. Discussion

Move the orca example further down, where more detail can be provided to understand the evidence.

After our extensive edits in the discussion, we felt this example was interrupting the flow. We now cite this study in the introduction.

Comment 9. Size dimorphism and evident sex differences are not discussed.

The revised manuscript now includes a new paragraph in the discussion in which sex differences are discussed (lines 319 – 339).

Comment 10. Be more precise in the terminology used (for example, land use seems to be interchangeable with habitat characteristics?).

We modified “land use” with “habitat data” in the revised manuscript.

Comment 11. Methods

Please provide a justification for the very high weight limit (5%; line 256). This limit is outdated and does not fulfill the international standard of 3% body weight. I assume the ethics clearance went through because of the short nature of the study (i.e., the birds were not burdened for life with the excess weight? But a line is needed here or under the ethics considerations to clarify this).

The 5% weight limit was considered acceptable due to the short deployment period, and we now edited the ethics statement to emphasize this point. However, it is important to note that there is no real international standard, with both 3% and 5% weight limits being commonly used. Both limits are arbitrary and the impact of a fixed mass on a bird varies with species and flight style. All owls survived and bred similarly to the non-tagged individuals in the population (lines 373 – 376 & lines 558 – 561)

EDITORIAL COMMENT: We strongly encourage you to provide further context and clarification on this issue, as suggested by the Reviewer. On a related point, the ethics

statement refers to GPS loggers, rather than GPS and ACC devices; we encourage you to clarify wording here.

Thank you for highlighting this point that indeed needed some clarifications.

Although we have used the terminology "GPS recorders", the authorization granted by the Swiss authorities for this study effectively covers the entire tracking system, which combines both GPS and ACC recorders in the same device. We have therefore changed the wording used in the ethics statement to avoid any misunderstanding (lines 373 – 376 & lines 558 – 561)

Comment 12. Please provide more information on the model selection approach, what does "Non-significant terms were dropped via model simplification by comparing model AIC with and without terms." mean? Did the authors use a stepwise backward elimination procedure (drop1 function)? Or did they apply a complete comparison of several candidate models? I think a model comparison approach rather than stepwise selection would be more informative, as several rather than only one model could be equally probable. This might also improve model weights or might require a model averaging procedure - current reported R2 values are very small and do not seem to support the results well.

We apologize for the lack of details about this important aspect of the statistical analysis. We applied an automated stepwise selection using the *dredge* function from the R package "MuMin", therefore applying a complete comparison of several candidate models. The final models were chosen as the best models since the number of candidate models within $\Delta AIC < 2$ was relatively low in each analysis and thus a model averaging was not appropriate here. We edited the methods section to ensure clarity, and added model selection tables for each analysis, ranked according to AICc scores, in the supplementary materials (lines 532 – 552)

In addition, we agree that the reported R-squared values in our analyses are quite low, specifically regarding the influence of pre-hunt perching force on hunting success (cond $R^2 = 0.04$). Nonetheless, landing impact still has a notable effect size (an increase of 1N reduces hunting success by 15%). The reported values are indicative of the inherent complexity in studying hunting behavior in a wild setting where numerous variables come into play. We specifically investigated the hypothesis that the force involved during pre-hunt landings, and consequently the emitted noise, influences the success of the next hunting attempt in wild barn owls. Factors such as prey behavior and micro-habitat characteristics surrounding prey (such as substrate type and vegetation height) are most likely to be influential but hard, or nearly impossible, to model. We now cover this in a more nuanced way in the discussion (lines 266 – 268)

Comment 13. Please explain why BirdID was nested in NightID - this is not clear to me.

Probably here there is a misunderstanding because we wrote that we nested NightID in BirdID (and not BirdID in NightID).

Comment 14. I hope the final graphs and legends will be larger, they are almost impossible to read.

We enlarged the graphs and legends as much as possible to improve readability. However, looking at the graphs in the published version they seem clear and readable.

Comment 15. Figure S1: Does "representation" mean the tracks don't show all of the 163 owls? If so, be precise and tell us how many are illustrated in the figure.

Figure S1 represent the tracks for each of the 163 barn owls used in the study. We changed the terminology used in the figure caption to avoid any misunderstanding.

| *Comment 16. Figure S4: Please adjust the y-axis to a readable format.*

Done

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